MILITARY SPECIFICATION

TELEPHONE SYSTEM, SOUND-POWERED, SHIPS SERVICE AN/WTC-2( )(V)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a ships service sound-powered telephone system, (circuit 1MJ) for Naval ships. The telephone system for any given installation will consist of varying quantities of table I equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-1842 - Cushioning Material, Plastic, Open Cell (For Packaging Applications).

MILITARY

MIL-R-19 - Resistors, Variable, Wire-Wound (Low Operating Temperatures), General Specification for.
MIL-R-94 - Resistors, Variable, Composition, General Specification for.
MIL-P-116 - Preservation, Methods of.
MIL-B-117 - Bags, Sleeves and Tubing - Interior Packaging.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5965

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MILITARY (Continued)

MIL-S-3950 - Switches, Toggle, Environmentally Sealed, General Specifications for.
MIL-B-5423 - Boots, Dust and Water Seal (for Toggle and Push-Button Switches, Circuit Breakers, and Rotary-Actuated Parts), General Specification for.
MIL-R-5757 - Relays, Electromagnetic, General Specification for.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-A-15303 - Audible Signals; Alarms, Bells, Buzzers, Horns, Sirens, and Electronic Shipboard.
MIL-T-15514 - Telephone Equipment, Sound Powered Handsets, Headset-Chest Sets, and Headset-Microphones, Types H-200/U, H-201/U, H-202/U, H-203/U, and H-204/U.
MIL-F-16377 - Fixtures Lighting: and Associated Parts; Shipboard Use, General Specification for.
MIL-F-16377/27 - Fixtures, Incandescent, General Lighting, 50 Watts, 120 Volts, Watertight, Symbols 92.2 and 93.2.
MIL-E-16400 - Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification for.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-C-21097 - Connectors, Electrical, Printed Wiring Board, General Purpose General Specification for.
MIL-S-22710 - Switches, Rotary (Printed Circuit), (Thumbwheel, Inline, and Pushbutton), General Specification for.
MIL-S-22710/12 - Switches, Rotary (Printed Circuit) (Thumbwheel Switch Assembly, Sealed Modules, Illuminated or Nonilluminated), (Top Relampable), Style SRPC12.
MIL-E-24142 - Enclosures for Electrical Fittings and Fixtures, General Specification for.
MIL-E-24142/2 - Enclosure for Electrical Fittings and Fixtures, Submersible (15-FOOT), Size 3 By 5.
MIL-S-24235 - Stuffing Tubes, Metal and Packing Assemblies for Electric Cables, General Specification for.
MIL-S-24235/14 - Stuffing Tube for Sheet Metal Enclosures.
MIL-T-55164 - Terminal Boards, Molded, Barrier, Screw and Stud Types and Associated Accessories, General Specification for.
MIL-B-81705 - Barrier Materials, Flexible, Electrostatic-Free, Heat Sealable.
MIL-P-81997 - Pouches, Cushioned, Flexible, Electrostatic-Free Recloseable, Transparent.

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STANDARDS

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-781 - Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
DOD-STD-1686 - Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices).

HANDBOOKS

MILITARY

DOD-HDBK-263 - Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices.) (Metric)

2.1.2 Other Government drawings and publications. The following other Government drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSHIPS)

S6501-73801 - Call Signal Station, Type IC/D.
S6501-74120 - Jack Sound Powered Telephone, Panel Mounted Type H-27A.
S6501-74210 - Jack Box Assembly Sound-Powered.
S6501-74464 - Handset Holder for Sound Powered Telephone Type Z-33A (Nonmagnetic).

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

SE003-AA-TRN-010 - Electrostatic Discharge Training Manual (S/N0910-LP-031-7300)

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publication. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI S1.4 - Sound Level Meters

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 General description. The system shall consist of equipment specified in table I. The equipment quantities and types required shall be as ordered (see 6.2.1).

3.2.1 System description. A simplified system configuration is shown on figure 1. The system shall be as required for interior shipboard use and shall have the following operating features:

(a) Provision for selection of from 1 to 144 call signal stations (including own station), from any station.
(b) Selective talking and ringing.
(c) Handset and headset-chestset operation.
(d) Ringing and talking on the same cable pair.
(e) Extension signaling both ringing and visual using extension relay jack boxes.
(f) Conference calls.
(g) Net operation.

3.2.2 System equipment. The system shall consist of the equipment as specified in table I. The equipment quantities and types to comprise a system shall be as specified by the contracting activity (see 6.2.1). A simplified system configuration is shown on figure 1. The system shall operate without the use of external power, except for extension signal devices.

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TABLE I. System equipment.

Item	Description paragraph	Requirement paragraph
Call signal station, side-crank bulkhead mounted	3.3.1	3.6.1
Call signal station, front-crank panel mounted	3.3.1	3.6.1
Call signal station test set	3.3.2	3.6.2
Audible alarm	3.4.1	3.6.3
Extension signal relay box	3.4.2	3.6.4
Extension signal device		
Visual (call lamp)	3.4.3.1	3.7
Acoustical (howler)	3.4.3.2	3.7
Extension phone jackboxes	3.4.4	3.6.5
Headset-chestset	3.4.5	3.8.3

3.2.3 Line sharing principle. To provide a maximum 144 terminal station capability, using a cable with only 12 active wire pairs, the equipment design shall provide that the individual lines (wires) in the twisted pair system interconnecting cable be shared among the stations. In each pair, one line leads to a specific tens digit switch position and the other leads to a matching position on the units digit switch (see table II and figure 2).

3.2.3.1 A six station simplified depiction of the line sharing principle is shown on figure 2. Stations whose pair begins with 7 (70, 71, 72...) share the tens digit line of pair 7; those whose number begins with 8 share the tens digit line of pair 8. Additionally, stations whose number ends with a 2 (72, 82...) share the units digit line of pair 2; those ending in 3 (73, 83...) share the units digit line of pair 3.

3.2.3.2 Correlation between cable pair, terminal station terminal board connections and stations selector switch connections shall be as shown in table II. For example, a call made to station number 11 (tens 1, units 1) shall utilize cable pair number 1, which is connected to terminals 1 and 1W within the terminal stations.

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TABLE II. Call signal station cable connections.

System interconnecting cable pair number	Terminal station terminal board connections		Station selector switch position	
	Black wire	White wire	Tens	Units
1	1	1W	1	1
2	2	2W	2	2
3	3	3W	3	3
4	4	4W	4	4
5	5	5W	5	5
6	6	6W	6	6
7	7	7W	7	7
8	8	8W	8	8
9	9	9W	9	9
10	0	0W	0	0
11	A	AW	A	A
12	B	BW	B	B

3.2.4 Looping limitation and terminal station number assignment. The system shall minimize looping by use of the number assignment tables. Looping is the unintentional formation of a complex series-parallel circuit among certain stations that are simultaneously busy, as shown on figures 3 and 4.

3.2.4.1 Number assignments for basic station-to-station telephone systems. A station-to-station call is one between call signal stations as selected. In order to optimize system performance the selection of station numbers shall be in accordance with table III. The selection process, in accordance with table III, is to begin the assignment with the left-most column and select numbers in the order of listing from top to bottom. Those stations which are expected to receive the greatest number of calls should be assigned the numbers in the left-most column. When the numbers in the left-most column have been assigned, the next column to the right is used.

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3.2.4.2 Number assignment for integrated station-to-station net systems. The integrated station-to-station and net systems are those in which certain system numbers are reserved for net (string circuit nets are specified in 3.2.6) communication (see figure 5). The numbers assigned to the telephone terminal stations in an integrated system shall be selected in accordance with 3.2.3.1, except that the selection shall be made in accordance with tables IV through IX. The numbers to be assigned for net access are also as specified in these tables. As an example, in planning an integrated system consisting of 29 stations and 4 nets, the number assignments are made in accordance with table VII. The four nets are assigned numbers 88, 99, AA and BB. The eight busiest stations (in terms of station-to-station calls received) are then assigned the numbers 00, 11, 22, 33, 44, 55, 66 and 77. The remaining 21 stations are then assigned 01, 12, 23..., 25, 36 and 47. In the event that additional stations are added later, they shall be assigned the numbers 50, 61...

TABLE III. Sequence for assignment of station numbers for basic station-to-station telephone systems. 1/

START	00	01	0A	03	05	08	04	09	07	06	02	0B
	11	12	1B	14	16	19	15	1A	18	17	13	10
	22	23	20	25	27	2A	26	2B	29	28	24	21
	33	34	31	36	38	3B	37	30	3A	39	35	32
	44	45	42	47	49	40	48	41	4B	4A	46	43
	55	56	53	58	5A	51	59	52	50	5B	57	54
	66	67	64	69	6B	62	6A	63	61	60	68	65
	77	78	75	7A	70	73	7B	74	72	71	79	76
	88	89	86	8B	81	84	80	85	83	82	8A	87
	99	9A	97	90	92	95	91	96	94	93	9B	98
	AA	AB	A8	A1	A3	A6	A2	A7	A5	A4	A0	A9
	BB	B0	B9	B2	B4	B7	B3	B8	B6	B5	B1	BA

1/ No nets.

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TABLE IV. Sequence for assignment of station and net number (systems with 1 net and up to 121 stations). 1/

START	00	01	20	25	38	51	59	74	83	93	A0
	11	12	31	36	49	62	6A	85	94	A4	10
	22	23	42	47	5A	73	80	96	A5	02	21
	33	34	53	58	70	84	91	A7	06	13	32
	44	45	64	69	81	95	A2	07	17	24	43
	55	56	75	7A	92	A6	09	18	28	35	54
	66	67	86	90	A3	04	1A	29	39	46	65
	77	78	97	A1	08	15	30	3A	4A	57	76
	88	89	A8	05	19	26	41	50	60	68	87
	99	9A	03	16	2A	37	52	61	71	79	98
	AA	0A	14	27	40	48	63	72	82	8A	A9

1/ Net number: BB

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TABLE V. Sequence for assignment of station and net numbers (systems with 2 nets and up to 100 stations). 1/

START	00	01	08	03	04	07	06	05	02	09
	11	12	19	14	15	18	17	16	13	10
	22	23	20	25	26	29	28	27	24	21
	33	34	31	36	37	30	39	38	35	32
	44	45	42	47	48	41	40	49	46	43
	55	56	53	58	59	52	51	50	57	54
	66	67	64	69	60	63	62	61	68	65
	77	78	75	70	71	74	73	72	79	76
	88	89	86	81	82	85	84	83	80	87
	99	90	97	92	93	96	95	94	91	98

1/ Net numbers: AA, BB

TABLE VI. Sequence for assignment of station and net numbers (systems with 3 nets and up to 81 stations). 1/

START	00	01	08	03	04	07	06	05	02
	11	12	10	14	15	18	17	16	13
	22	23	21	25	26	20	28	27	24
	33	34	32	36	37	31	30	38	35
	44	45	43	47	48	42	41	40	46
	55	56	54	58	50	53	52	51	57
	66	67	65	60	61	64	63	62	68
	77	78	76	71	72	75	74	73	70
	88	80	87	82	83	86	85	84	81

1/ Net numbers: 99, AA, BB

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TABLE VII. Sequence for assignment of station and net numbers (systems with 4 nets and up to 64 stations). 1/

START	00	01	06	03	05	04	02	07
	11	12	17	14	16	15	13	10
	22	23	20	25	27	26	24	21
	33	34	31	36	30	37	35	32
	44	45	42	47	41	40	46	43
	55	56	53	50	52	51	57	54
	66	67	64	61	63	62	60	65
	77	70	75	72	74	73	71	76

1/ Net numbers: 88, 99, AA, BB

TABLE VIII. Sequence for assignment of station and net numbers (systems with 5 nets and up to 49 stations). 1/

START	00	01	06	03	05	04	02
	11	12	10	14	16	15	13
	22	23	21	25	20	26	24
	33	34	32	36	31	30	35
	44	45	43	40	42	41	46
	55	56	54	51	53	52	50
	66	60	65	62	64	63	61

1/ Net numbers: 77, 88, AA, BB

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TABLE IX. Sequence for assignment of station and net numbers (systems with 6 nets and up to 36 stations). 1/

START	00	01	04	03	02	05
	11	12	15	14	13	10
	22	23	20	25	25	21
	33	34	31	30	35	32
	44	45	42	41	40	43
	55	50	53	52	51	54

1/ Net numbers: 66, 77, 88, 99, AA, BB

3.2.5 Operating procedures for call signal stations (see figure 6). Operation of the call signal stations and handsets shall be as specified in 3.2.5.1 through 3.2.5.3.

3.2.5.1 Calling procedure.

(a) Set the call-answer switch to the call position. (If the switch is left in the answer position, unintentional self-ringing will occur at the calling station when crank is turned and ringing will not occur at the called station.)
(b) Set the dials of the station selector switch to the desired number.
(c) Depress handset push-to-operate switch, and listen for talking to determine if the selected station is busy or not.
(d) If not busy, rotate the magneto generator crank through several turns to ring the called station.
(e) When the called station answers, depress the push-to-operate button and proceed to talk over the handset. Readily intelligible conversation is required.

3.2.5.2 Call answering procedure. Place the call-answer switch in the answer position and press the handset press-to-operate button. (A called station will still ring even if its call-answer switch has been left in the call position; however, no communication will be possible until the switch is placed in the answer position.)

3.2.5.3 Conference call procedure. Conference call members shall call one station and that station calls itself. Four members shall not cause more than 50 percent reduction in signal, when tested in accordance with 4.7.20.

3.2.6 Nets. Net circuits are two wire branches taken off a station-to-station circuit (see figure 5). The branches connect to jack boxes for handsets and headset/chestsets. Nets are given exclusive wire pairs (as was done in tables IV through IX) because they are more likely to be in use and therefore line sharing would not be as practical. The call signal station headset/chestset jack is usually wired directly into the net so that it may be

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used while the station is making station-to-station calls with the handset. The station handset may also be used in the net circuit but the net circuit must be selected. Net branch equipment cannot be rung.

3.3 Basic system components.

3.3.1 Call stations, front-crank TA-975/WTC-2(V) and side-crank TA-974/WTC-2(V). This equipment shall provide a means of selecting and calling other stations and of answering calls from other stations in the system. The H-203/U sound-powered handset and the holder (Z-33A) shall be supplied as part of the terminal stations (see figures 7 and 8).

3.3.2 Call signal station test set TS-3687/WTC-2(V). This portable equipment shall provide a means to quickly test the terminal selector switch and associated circuitry (see figure 9).

3.4 Ancillary device description.

3.4.1 Audible (extension howler) BZ-240/WTC-2(V). This equipment shall provide an audible ring signal at a location remote from the terminal station (see figure 10).

3.4.2 Extension signal relay box J-3523/WTC-2(V). This device shall consist of an enclosure housing a call signal-activated relay, which itself activates one or more externally powered bells, buzzers or lamps. This device shall be used in conjunction with terminal stations located in high noise areas (see figure 11).

3.4.3 Extension signal devices. The following devices are used in the AN/WTC-2 sound powered telephone system.

3.4.3.1 Visual signal device. This device shall provide a visual signal to indicate an incoming call in areas where high noise levels exist. This device shall be in accordance with MIL-F-16377 and MIL-F-16377/27. It shall be operated in accordance with 3.4.2.

3.4.3.2 Acoustical signal device. This device in accordance with MIL-A-15303, shall provide an acoustical signal to indicate an incoming call in areas where high noise levels exist.

3.4.4 Extension phone jackboxes (see S6501-74210). This equipment shall provide connection for headset-chestset, type H-200/U or handset H-203/U into a sound powered telephone (SPT) circuit at some location remote to the two interconnected call stations.

3.4.5 Headset-chestset, type H-200/U (see MIL-T-15514). This general purpose sound-powered headset-chestset shall be used with the H-27A jack installed in the terminal stations and extension jackboxes.

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3.5. General requirements. The following provides additional specifications to MIL-E-16400 requirements (refer to MIL-E-16400, 6.2.1):

(a) Mercury, mercury compounds and radioactive compounds shall not be used in the equipment or the manufacturing process.
(b) Asbestos materials shall not be used in the equipment.
(c) Temperature range shall be range 2.
(d) Normal operation when inclined 60 degrees is required. MIL-E-16400 testing is not required. Verification testing in accordance with an approved test procedure is required.
(e) Fuse protection is not required.
(f) Dielectric withstanding voltage testing shall be between adjacent call signal lines and to ground at the selector switches.

3.6 Mechanical and electrical requirements of sound-powered telephone systems AN/WTC-2. AN/WTC-2 sound-powered telephone systems shall be in accordance with 3.6.1 through 3.6.5.

3.6.1 Call signal stations. Call signal stations shall be contained in watertight enclosures and shall be in accordance with 3.6.1.1 through 3.6.1.6.9.

3.6.1.1 Mounted assembly. Standard cable entrance stuffing tubes and terminal blocks along with a handset connector and a headset-chestset connector shall be required (type H-27A). The terminal block shall be wired into a connector receptacle for joining to the front face assembly. Terminal blocks shall be in accordance with MIL-T-55164.

3.6.1.2 Front cover assembly. The front cover assembly shall mount the magneto generator gear and crank shaft subassembly and howler in a separate watertight enclosure. Should this subassembly leak, water shall be contained from reaching the main enclosure or the other subassemblies. The front cover assembly shall include all controls and electronics. The front face assembly shall attach to the mounted assembly by retained hex head bolts having screwdriver slots. It shall be removable for maintenance without any action on the mounted assembly.

3.6.1.3 Handset and holder assembly. Unless otherwise specified (see 6.2.1), each call signal station shall be provided with one H-203/U handset, prewired and one handset holder, in accordance with Drawing S6501-74464. H-203/U handsets shall be in accordance with MIL-T-15514.

3.6.1.4 Dimensions. The call signal station maximum dimensions exclusive of the crank and crankshaft shall be 8-1/2 inches high by 6-1/2 inches wide by 6-1/2 inches deep. The addition of the crank and crankshaft shall not increase any overall dimensions by more than 2 inches. Mounting dimensions and shape shall conform to Drawing S6501-73801.

3.6.1.5 Enclosure material. Enclosures shall be fabricated from cast aluminum alloy (see 3.5). Minimum thickness of housing material shall be 0.16 inch.

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3.6.1.6 Electrical controls and components. Call signal stations shall be provided with a self-ringing capability. They shall be protected against cable shorts and reverse polarity signals. Simultaneous calls to one station shall not cause change in performance of a two-station call. Only the called station shall ring when called.

3.6.1.6.1 Station selector switch. The station selector switch shall consist of two single pole, 12 position, bi-directional, no stops, thumbwheel actuated switches in accordance with MIL-S-22710 and MIL-S-22710/12. Each switch shall have a drum type indicator numbered from 0 through 9 with the remaining two positions labeled A and B. The switch seal shall have been tested in accordance with method II of MIL-S-22710. Negative air pressure is not required. Connectors in accordance with MIL-C-21097 may be used with this switch. The enclosure test (see 4.7.18) shall not cause change in resistance across the closed contacts or to ground from the closed contacts greater than plus or minus 5 percent, nor shall the same resistances change more than plus or minus 5 percent due to the accelerated life test specified in 4.7.22.

3.6.1.6.2 Generator assembly. The generator assembly shall consist of the generator, gears, crankshaft, crank, necessary wire and crankshaft penetrations, seals, and attachment hardware. The generator output shall be as required to power two howlers and one extension relay at 12 volts direct current (Vd.c.) after 40 ohms line resistance and shall be in accordance with 3.6.1.6.4. This shall be achieved turning a 1.4 inch or less radius crank at 2 to 2.5 revolutions per second (r/s). Less than 7 inch-pounds of torque shall be required to maintain this rate of revolution. A centrifugal or electronic rate switch shall be used so that output is not applied until a usable voltage level is reached. Plus 12 Vd.c. shall be on the ten terminal when measured with respect to the units terminal.

3.6.1.6.2.1 Generator output short circuit protection. After testing in accordance with 4.7.5, the howler output shall meet the requirements of 3.6.1.6.4 and 3.6.1.6.4.1.

3.6.1.6.2.2 Generator noise. The generator 12 Vd.c. output shall not introduce noise into the handset receiver or the headset of the headset-chestset that is above normal conversation level (see 3.8.3). The generator alternating current (a.c.) harmonics, that will pass blocking capacitor (see 3.8.3) shall not be high enough to appear to be loud over a handset or a headset when the button is depressed during cranking of the generator and output of generator signal on the same line.

3.6.1.6.3 Call-answer switch. A two-position toggle type switch shall be provided. The up position shall be labeled "PLACE CALL" or "CALL" and the down position labeled "ANSWER." A double (or more) pole, double throw, switch in accordance with MIL-S-3950 shall be used with a boot in accordance with MIL-B-5423.

3.6.1.6.4 Howler sound characteristics. When tested in accordance with 4.7.7 and 4.7.8, each howler audio output shall be at least 90 decibel-A response (dBA) sound pressure level at a distance of 2 feet.

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3.6.1.6.4.1 Frequency. The howler audible output shall be a single tone varied from 800 ± 80 to 1360 ± 140 hertz (Hz) and return, using a logarithmic frequency change, calibrated in decades per unit of time. The frequency change cycle shall occur at 5 ± 0.5 Hz and at 10 ± 1 Hz as selected internally by a two-position switch or shorting bar. The purpose of these selectable warble rates is to provide a different tone for two call signal stations that are located in the same area.

3.6.1.6.5 Transients. When tested in accordance with 4.7.9, switching clicks and transients shall not cause audible noise in a handset or headset and shall not cause extension relay operation.

3.6.1.6.6 Volume control. An internal volume control for own station howler and another volume control for own station extension howler shall be provided. A sound pressure output range adjustment of 90 to 50 decibel (dB) on the A scale (in accordance with ANSI S1.4, type 2, A weighting for sound level meters) shall be provided in 6 dB steps or in a continuous manner. Variable resistors shall be in accordance with MIL-R-19 or MIL-R-94, as applicable.

3.6.1.6.7 Privacy and noise. Any howler self generative signals shall be blocked from introduction on the audio circuits. Both lines must be open (1 megohm) unless a certain threshold of voltage is present such as might be present for an incoming call signal from another station.

3.6.1.6.8 Wiring. The call signal station shall be connected in such a manner that when station selector switches are dialed to positions 1-1, 2-2, 3-3, ---, 9-9, 0-0, A-A, B-B that cables pairs 1, 2, 3, --, 9, 10, 11, 12 are selected respectively (see table II).

3.6.1.6.8.1 Terminal board connections. Terminal board jumpers for both station number determination and H-200/U headset-chestset jack connection shall be pre-installed at one end, with the other end terminated in a lug but left unconnected. Jumper wires shall be of a length to allow any station to be wired as any system number, and to allow the jack to be connected to the handset terminals or to any wire pair. Terminals designated H+ and H- shall be provided for incoming calls and for connection of own station number jumpers (to allow calling own station). Terminals designated P+ and P- shall be provided at the telephone terminals station ships wiring terminal board for connecting the station handset. Terminals designated R+ and R- shall be provided at the telephone terminals station ships wiring terminal board for connecting the coil of the extension signal relay. Terminals designated E1 and E2 shall be provided for connection of an extension howler (see figure 12).

3.6.1.6.9 Alternate crank location. Call signal stations shall be provided with crank located at the front or side as specified (see 6.2.1). Both types shall have successfully completed first article inspection testing (see figures 13 and 14).

3.6.1.7 Labels and plates. Labels and plates in accordance with MIL-P-15024 and MIL-P-15024/5, except type G (adhesive-backed metal foil), shall be acceptable for tagging, labeling and instructions. Location and relative letter size to be used shall be as shown on figures 13 and 14. Lettering shall be engraved or embossed.

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3.6.1.7.1 Information plate. The information plate shall provide the station number and operating instructions as shown on figures 13 and 14. The statement "EXCESSIVE TRAFFIC MAY CAUSE LINE SHARING INTERFERENCE" shall be added.

3.6.1.7.2 Labels. Label plates shall be provided for the call-answer switch and the station select switches.

3.6.2 Call signal station test set. A portable test set shall be provided to allow quick continuity testing of the station selector switches and associated circuits (see figure 15). The test set shall have a connector that mates to the call stations' cover connector. The test set shall be energized with 115 volts (V) at 60 Hz. Maximum dimensions are 15 by 12-inch face, 10-inch depth. Maximum weight is 10 pounds.

3.6.2.1 Indicators. The test set shall have two rows of light emitting diodes (LEDs), one for each switch, consisting of 12 LEDs to correspond to the 12 switch positions. As the switch positions are changed, the corresponding LED shall glow to signify switch and associated circuit continuity, and any possible shorts, when more than one LED in each row glows. The use of commercial LEDs is acceptable for the test set.

3.6.2.2 Self-test feature. The test set shall include a built-in LED self-test feature to ensure proper circuit operation.

3.6.3 Extension howler. The calling station shall drive two howlers at the called station. One of these is an extension howler paralleled to operate with the station howler. The extension howler shall be identical to the station howler except it shall be enclosed in a watertight (15 foot) brass or aluminum enclosure for bulkhead mounting. When tested in accordance with 4.7.7 and 4.7.8, the extension howler output shall be in accordance with 3.6.1.6.4. The overall volume shall be less than 0.25 cubic feet, maximum weight is 2.1 pounds.

3.6.4 Extension signal relay. The extension signal relay shall be mounted in a submersible (15 foot) brass or aluminum bulkhead mounted enclosure in accordance with MIL-E-24142 and MIL-E-24142/2 (see figure 11 for assembly and figure 16 for schematic). Maximum volume allowed is 0.35 cubic feet. Maximum weight allowed is 3.5 pounds.

3.6.4.1 Relay. The relay station shall operate at 2 or less crank r/s of the generator when calling a station having a howler and extension howler. Relay drop out voltage shall be at least 20 percent below pick up voltage. Contact rating shall be at least 300 Vd.c., 3 amperes (A). Two single pole, single throw contacts are required. Drop out time delay shall be at least 1.5 seconds after cranking stops. When tested in accordance with 4.7.17, the test contact resistance shall not have increased more than 10 percent. Other relay requirements shall be in accordance with MIL-R-5757.

3.6.5 Extension jackboxes. Extension jackboxes shall be in accordance with Drawing S6501-74210. Type H-27A jacks in accordance with Drawing S6501-74120 shall be used and water tightness shall be maintained.

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3.7 Powered devices. The following devices may be operated by the extension signal relay to apply ships service power:

(a) Audible signals, alarms, bells, buzzers, horns and sirens in accordance with MIL-A-15303.
(b) Fixtures, lighting and associated parts in accordance with MIL-F-16377/27.

3.8 Protective circuit requirements. The following operating conditions as specified in 3.8.1 through 3.8.4 shall not cause any performance deterioration or failures.

3.8.1 Simultaneous call protection. When two or more stations call at the same station simultaneously, no damage to the called station shall result.

3.8.2 Reverse signaling protection. Inadvertent reversal of signaling voltage shall not damage any system equipment.

3.8.3 Handset and headset protection. Circuitry shall be provided to assure that no ear damage or equipment damage will result if the call signal voltage is applied to the handset or headset when the push-to-operate switch is operated. No sound above 120 dB and no sound above 90 dB for 2 seconds referred to 20 micro pascales shall occur at the earpiece, that is generated within this equipment. A d.c. decoupling capacitor circuit shall be used. Audio frequency above 195 Hz shall not drop more than 2 decibel volts (dBV) across this capacitor.

3.8.3.1 Handset audio level. When tested in accordance with 4.7.21, the reduction in audio level of sound transmitted via the H-203/U handset, due to loading of the audio circuit by the terminal station signaling circuitry, shall be less than 1 dB.

3.8.4 False ring. Looping (see figure 4) shall not cause false ringing. When tested in accordance with 4.7.12, only the called station shall ring.

3.9 Environmental requirements. Specific equipments shall be subjected to environmental requirements in accordance with table X. Unless otherwise specified (see 6.2.1), the equipment shall be energized and operating at rated signal voltage and load current. Simulated inputs may be used. Performance requirements shall be met and tested during all environmental exposures and after exposure, except during shock.

TABLE X. Environmental requirements.

Environmental exposure requirements	Item		
	Terminal station	Extension signal relay box	Extension howler
Salt fog	X	X	X
Shock, vibration inclination verification	X	X	X
Watertight enclosure - 15 feet	X	X	X
Accelerated life	X	X	

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3.9.1 Shock. When tested in accordance with 4.4, equipment shock and post shock performance requirements shall be within the specified limits in accordance with 3.5.

3.9.2 Vibration. Equipment vibration requirements shall be in accordance with 3.5. The required upper vibration frequency is 50 Hz. Performance during and after vibration shall be within the specified limits in accordance with 3.5 and as specified herein.

3.9.3 Inclination. Equipment inclination requirements shall be in accordance with 3.5. Performance during and after 60-degree inclination tests shall be normal.

3.9.4 Enclosure. The various system components shall meet the requirements for enclosure in accordance with table X. The watertight enclosure test shall be in accordance with 4.7.18. The call signal stations shall not leak water into any internal compartment during operation of the switches, the howler or the generator crank. Cable entrance shall be in accordance with MIL-S-24235 and MIL-S-24235/14.

3.9.5 Salt fog. After being tested in accordance with 3.5 for sheltered equipment, this equipment shall show no deterioration of coatings and surfaces. Post test performance and examination shall be in accordance with 4.4 and as specified herein.

3.9.6 Accelerated life. The call signal station shall withstand the test specified in 4.7.22 without mechanical or electrical damage, adjustments or deterioration in performance.

3.10 Reliability and maintainability (R/M) program. When specified, the contractor shall prepare a R/M program plan in accordance with the data ordering document (see 6.2.2).

3.10.1 Mean-time-between-failure (MTBF) requirement. The MTBF as defined in MIL-STD-781 for the equipment specified herein shall not be less than 50,000 hours when demonstrated in accordance with 4.7.22.3. Failure definition shall be in accordance with MIL-STD-781.

3.10.2 Preventative maintenance requirements. The equipment shall not require preventative maintenance for conformance to 3.10.1. Preventative maintenance is recommended but not required for the purpose of increasing MTBF or life of the equipment.

3.10.3 Calibration, alinement and adjustment requirement. The equipment shall not require calibrations, alinements or adjustments at the organizational level, other than setting of howler level and tone.

3.10.4 MTBF demonstration. Failure to meet the MTBF requirements of 3.10.1 during accelerated life testing demonstrates noncompliance and such equipment shall not be offered for delivery.

3.11 Technical manuals. The contractor shall prepare manuals in accordance with the data ordering documents (see 6.2.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system in accordance with the data ordering documents included in the contract or order (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified herein, all measurements and tests shall be made under the following standard ambient conditions:

(a) Temperature: 73 ± 18°F.
(b) Relative humidity: 50 ± 30 percent.
(c) Atmospheric pressure: 28.5 plus 2.0 minus 3.0 inches of mercury.

4.4 First article inspection. First article inspection shall consist of the examination and tests in accordance with table XI. The sample items shall be connected as a system for this test, adding 40 ohms line resistance between stations. The tests shall be conducted in the order specified in table XI, without unreported maintenance or adjustments. Maintenance must be reported to the contracting activity and the Government inspector when performed. Replacement of parts and assemblies shall be justification for retesting to the extent required by the Government.

4.4.1 First article test procedure. The first article test procedure shall be in accordance with the data ordering documents (see 6.2.2).

4.4.1.1 Notification of tests. Notification of tests shall be in accordance with the data ordering documents (see 6.2.2).

4.4.2 First article inspection report. A first article inspection report shall be prepared in accordance with the data ordering documents as specified in the contract or order (see 6.2.2).

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TABLE XI. First article inspection.

Required sample: 3 call signal stations (2 side and one front crank) 2 extension howlers, 2 extension relays.

System connection: Connect 2 stations with extension howlers and extension relays and one station with no extension devices. Interconnect all 3 stations for normal calling.

Test order	Inspection	Requirement paragraphs	Test paragraphs
1	Operating procedures	3.2.5	4.7.1
2	Howler output	3.6.1.6.4	4.7.7
3	Howler frequency	3.6.1.6.4.1	4.7.7, 4.7.8
4	Relay operation	3.6.4.1	4.7.17
5	Station selector switches	3.6.1.6.1	4.7.19, 4.7.3
6	Accelerated life test	3.9.6	4.7.22
7	Salt fog (sheltered equipment)	3.9.5	MIL-E-16400
8	Shock	3.9.1	MIL-E-16400
9	Vibration	3.9.2	MIL-E-16400
10	Dielectric withstanding voltage	3.5(f)	MIL-E-16400
11	Watertight enclosure - 15 ft	3.6.4, 3.9.4	4.7.18, table X
12	Operating procedures	3.2.5	4.7.1
13	General examination	3.5	4.7.1, MIL-E-16400
14	Electrical test using test set	3.6.2	4.7.3
15	Call signal station self test	3.2.5.3	4.7.2
16	Handset and headset protection	3.8.3	4.7.11
17	Conference call	3.2.5.3	4.7.20
18	Handset audio level	3.8.3.1	4.7.21
19	Generator output short circuit protection	3.6.1.6.2.1	4.7.5
20	Volume attenuator	3.6.1.6.6	4.7.10
21	Howler output (and extension howler)	3.6.1.6.4, 3.6.3	4.7.7
22	Howler frequency (and extension howler)	3.6.1.6.4.1, 3.6.3	4.7.7, 4.7.8
23	General examination	3.5	4.7.1, MIL-E-16400
24	Extension signal relay, operation	3.6.4.1	4.7.16, 4.7.17
25	Generator noise	3.6.1.6.2.2	4.7.11
26	Transients	3.6.1.6.5	4.7.9
27	Generator assembly test	3.6.1.6.2	4.7.6
28	Station selector switch	3.6.1.6.1	4.7.3
29	Reverse polarity	3.8.2	4.7.13, 4.7.4
30	False ring test	3.8.4	4.7.12

4.5 Sampling for quality conformance inspection. Sampling shall be performed in accordance with MIL-STD-105 for each item of the group under acquisition specified in table I, less headset/chestset and visual call lamp.

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4.5.1 Lot. Equipment produced under this specification, presented for delivery at one time, shall be considered a lot. For the purposes of inspection, all components of a single type shall be the basis for number of equipment in an inspection lot. The lot may include the entire contract quantity or it may be the production of any convenient quantity or frequency as chosen by the inspector.

4.5.2 Sampling for group A inspection. Each equipment of a lot shall undergo group A tests. Deficient equipments found by group A inspection must be corrected and successfully retested before other testing or delivery.

4.5.3 Sampling plan for group B inspections. The lot or batch shall be the initial contract quantity or 100, whichever is smaller. Additional quantities added by contract modification shall be a new lot. Equipment of each type that passed group A tests are required to be sampled regardless of lot size. The AQL for defective equipment shall be as specified in table XII for the double sampling plan for reduced inspection and tightened inspection in accordance with MIL-STD-105.

4.5.3.1 Rejected lots. Rejected lots shall not be offered for delivery until corrective action acceptable to the acquiring activity has been demonstrated and documented. One hundred percent correction and inspection of the rejected lot is required until the tightened inspection AQL has been reestablished. Group B tests not affected by the corrective action in the opinion of the acquiring activity need not be repeated.

4.5.4 Sampling for group C inspection. When the basic design of the equipment, or the material of a vital part has been changed, applicable group C test inspection is required.

4.6 Quality conformance inspection. Quality assurance terms and definitions shall be in accordance with MIL-STD-109.

4.6.1 Quality conformance inspection report. A quality conformance inspection report shall be prepared in accordance with the data ordering documents (see 6.2.2).

4.6.2 Group A. Equipment shall at least be subjected to the group A tests as specified in table XII. Failure to pass contractor production quality assurance inspections and failure to conform to this specification shall be counted as a defect and the equipment shall not be offered for delivery. All equipment under contract shall be corrected and reinspected.

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TABLE XII. Quality conformance inspections.

Set under test: Three or more call signal stations having extension devices as ordered.

Tests	Requirement paragraphs	Test paragraphs	AQL
Group A:			
General examination	3.5	4.7.1, MIL-E-16400	N/A
Electrical test using test set	3.6.1.6.1	4.7.3	N/A
Call signal station self test	3.2.5.3	4.7.2	N/A
D.c., signal circuit polarity	3.8.2	4.7.4, 4.7.13	N/A
Relay operation	3.6.4.1	4.7.17	N/A
Howler output	3.6.1.6.4	4.7.7	N/A
Contractor production control inspections	4.1	Contractor test procedures	N/A
Group B:			
Conference call	3.2.5.3	4.7.20	1.0
Short circuit protection	3.6.1.6.2.1	4.7.5	0.65
Enclosure submersion test	3.6.1, 3.6.4	4.7.6, 4.7.18	1.0
Howler assembly test	3.6.3	4.7.18.1	1.0
Extension signal relay	3.6.4	4.7.16	1.0
Group C:			
Group B tests			
Inspection tests from first article test acceptable to the acquiring activity		Table XII	

4.6.3 Group B. Equipment selected in accordance with 4.5.3 shall be subjected to the group B tests as specified in table XII. Failures and rejection shall be under the same requirements specified in 4.6.2 for group A tests.

4.6.4 Group C. Equipment shall be subjected to, but not limited by, group C tests as specified in 4.5.4 and table XII. The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any group C tests the contractor shall redesign and retest as required by the acquiring activity.

4.7 Tests. Sound pressure outputs shall be measured using an international electrotechnical commission sound level meter and A weighting network and slow response in accordance with ANSI S1.4.

4.7.1 General examination. Components of the equipment ordered shall be given a general examination for workmanship, to assure conformance with 3.5, and to assure the parts used were in accordance with 3.10. In addition, a demonstration in accordance with 3.2.5 with at least two stations shall be required. First article examination requires disassembly to the extent required for the inspector to examine all mechanical parts.

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4.7.2 Call signal station self test. The terminal station shall be temporarily wired so that it will ring as system station number 11, by connecting terminals H+ and H- to terminals 1 and 1W. With the station selector switches set to number 11, it shall be verified that the station rings when cranked, with the call-answer switch first in the call position, and then in the answer position. The station selector switches shall then be set to 22, and it shall be verified that the station will ring itself only when cranked in the answer mode.

4.7.3 Electrical test using test set. The terminal station shall be connected to a test set (see figure 15). Station numbers 00, 11, 22...99, AA, BB shall be set on the station selector switches, and it shall be verified that the corresponding LED indicators light in the test set, indicating proper wiring and station selector continuity.

4.7.4 D.c. signal circuit polarity test. The station selector shall be in the 11 position, the call-answer switch shall be in the call position, and the generator shall be cranked. The output voltage shall appear on the 11 terminals (cable pair number 1 (see figure 15)) and the tens terminal shall be 12 V positive with respect to the units terminal.

4.7.5 Short circuit protection. The d.c. output circuitry of the terminal shall be shorted, simulating a cable short. With the short in place, the set shall be cranked at 140 (or more) revolutions per minute (r/min) for 15 seconds. With the short removed, conduct howler output test in accordance with 4.7.7.

4.7.6 Generator assembly test. Two terminal stations with extension howlers and extension signal relays shall be configured so that each station is capable of calling the other. The calling station shall be cranked at 2.0 to 2.5 revolutions per second (r/s). Measure torque required to maintain this r/s. Operate the assembly under 15 feet of water (or 3 inches of water at 6.4 pounds per square inch (lb/in^2) differential pressure for 5 minutes of continuous rotation at 2 r/s). Inspection for leaks by the Government inspector shall be required. In accordance with MIL-STD-108 monitor differential pressure during the test.

4.7.7 Howler output. Using the configuration in accordance with 4.7.6 and 10 Hz warble, the howler outputs shall be measured with and without the extension relay. With both howlers and the relay energized by the calling station, the dBA sound pressure 2 feet from each howler shall be measured. Repeat with relay disconnected. The frequencies of the audible signal and the warble frequency shall be measured by a calibrated laboratory microphone and amplifier.

4.7.8 Warble selection. Repeat the test specified in 4.7.7 at 5 Hz warble rate.

4.7.9 Transient test. On an oscilloscope and by listening via a H200/U headset, observe the effects on the receiver of switching and calling by other stations in the test system. Set the oscilloscope amplifier gain to where the switching transients are clearly visible and measurable. When a common line is shared by the station under test and the other stations, the oscilloscope transient peak level caused by the above switching shall be below the audible level signal of the H200/U headset.

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4.7.10 Volume attenuation. When the terminal station is cranked at a steady 2.0 to 2.5 r/s and loaded in accordance with 4.7.6, the volume control shall be set to allow maximum output of the howler. The sound pressure output of the howler shall be measured at 2 feet distance in an anechoic chamber. The measurement with the attenuator set to lowest volume setting shall be repeated (see 3.6.1.6.6).

4.7.11 Handset-headset ear protection. The call station shall be set on its own number, and the generator shall be cranked at a steady 3.0 to 4.0 r/s, with the press-to-operate switch on the H203/U handset depressed, thus connecting the handset to the call signal voltage. Using a H200/U headset-chestset, the test shall be repeated. The sound pressure level at 1/4 to 1/2 inch from the reproducer shall be measured and recorded using a standard sound level meter. The receiver shall be coupled to the microphone in accordance with MIL-T-15514.

4.7.12 False ring test. Four terminal stations shall be wired together to form a system, with the stations having system numbers 54, 24, 26 and 56. These numbers are one set of numbers which are always electrically interconnected via a sneak circuit as a result of the cable wire-sharing principle inherent to this system. Using station 54, call station 56. It shall be verified that the called station rings and that the other stations do not ring. With the same station setup, use three stations to simultaneously call the fourth, continuously, for 15 seconds. Then self test each station to assure that each is still functioning properly.

4.7.13 Reverse polarity protection. An intentional wiring error shall be installed on the d.c. signal lines. One station shall call the other at 4 plus r/s for 15 seconds. No damage shall be incurred by the called station. Verify by correcting the wiring error, calling and using the talk test (see 3.8.2).

4.7.14 Call station test set operational test. The test set shall demonstrate ability to indicate an open circuit and a short circuit and the circuit they are in by altering a good call signal station to have these circuit defects.

4.7.15 Handset loading. Handset transmitter output shall be measured before and during connection to a call signal station while the same continuous audio tone is applied using a standard laboratory artificial voice audio generator.

4.7.16 Extension signal relay sensitivity. The pickup and drop out voltage level and r/s required at these voltage levels shall be measured before and immediately after 1 hour of application of rated coil voltage. Subsequent to this test and after the relay has returned to ambient room temperature the above measurements shall be repeated. The first test above shall be repeated at a stabilized ambient temperature of 149°F.

4.7.17 Extension signal relay contact resistance. An extension signal relay shall be connected to a call signal station. Connect 115 V, 60 Hz power via the relay contacts to three visual indicators in parallel. The relay coil with the call signal generator shall be energized, measure the contact resistances while cranking at 2 or 3 r/s with the above load disconnected. The load shall be reconnected and alternately energize the relay for 3 seconds ON and 2 seconds OFF. After 50 cycles the contact resistance, howler generator voltage and r/s required to close the normally open contacts shall be measured. Also, the drop out voltage drop out time and r/s shall be measured.

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4.7.18 Enclosure test. During the watertight enclosure test (see 3.6), controls shall be operated at least 10 times through all positions while the equipment assembly is submerged to test pressure. Leakage shall not be acceptable.

4.7.18.1 Howler mechanism. In accordance with MIL-STD-108 the crankshaft seal shall be tested by rotating the crank while submerged in 15 feet of water (or 3 inches of water and 6.4 lb/in^2 differential pressure) for at least 15 minutes continuously at 3 r/s. An absorbent cotton packing material shall be placed on the inside of the bearing seal to be used to absorb any moisture passage along the shaft and any moisture thrown out by the bearing. This material shall be weighed before and after this submergent test to detect leakage. Five percent or more increase in weight is leakage, lubricant absorbed is considered moisture. Disassembly and inspection for moisture shall be required. A dye in the water shall be required to aid such inspection. Leakage shall not be acceptable.

4.7.19 Station selection switch. Contact resistance shall be measured before and after accelerated life test and enclosure test for conformance to 3.6.1.6.1.

4.7.20 Conference call. Four call signal stations in conference call shall be simulated using the three signal stations and an equivalent impedance for the fourth station. The root mean square voltage signal level at the receivers shall be measured when connected to one station and to four stations. The percentage change in level shall be reported.

4.7.21 Handset audio level. The open circuit output 80 microbar of an H-203/U hand set shall be measured with a 1000 Hz sinusoidal continuous signal. The output shall be connected to a call signal station and repeat the measurement (see 3.6.1.3).

4.7.22 Accelerated life test. Three call signal stations shall be operated including ancillary equipment for 24,000 cycles. The call signal stations shall be configured with an extension relay box and a visual signaling device. Each station shall also be connected to an extension howler. One station shall be placed in the answer mode so that the two howlers, extension signal relay box and visual indicator can operate. A cycle shall consist of bringing the calling generator crank up to a specified r/s in not more than 2 seconds, that speed being maintained for 7 ± 2 seconds and stopping the generator crank in not more than 2 seconds. Before start of the next cycle, rotate the station selector switches at least 120 degrees. This test shall be conducted in accordance with table XIII; the test shall be continuous within each temperature setting shown. When tested in accordance with 4.7.3 and 4.7.7, conformance to 3.6.1.6.2 and 3.6.1.6.4 is required. During accelerated life testing an electronic signal equivalent to the 80 microbar signal (see 4.7.21) shall be applied.

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TABLE XIII. Accelerated life test.

Cycles	Crankshaft r/s	Ambient temp. °F (°C)	Relative humidity	Soak time 1/ hours	Test no.
0-7000	2-2.5	75 ± 3 (25 ± 1.5)	Not controlled	0	1
7001-12,000	2-2.5	149 ± 3 (65 ± 1.5)	95 percent	1	2
12,001-17,000	2-2.5	-20 ± 3 (-28 ± 1.5)	95 percent	2	3
17,001-22,000	2-2.5	149 ± 3 (65 ± 1.5)	5 percent	2	4
22,001-24,000	2-2.5	75 ± 3 (25 ± 1.5)	Not controlled	1	5

1/ Soak time = Time at the required ambient temperature before start of cycling.

4.7.22.1 Performance monitoring. Throughout the accelerated life test the following performance parameters shall be measured and recorded at the frequency indicated:

(a) Howler sound pressure output - at least every 200th cycle.
(b) Receiver (earpiece) soundpressure output - continuous dBA monitoring.
(c) Selector switches operation - continuous continuity monitoring.

4.7.22.2 Performance tests. The following performance tests shall be conducted after soak time but before start of cycling; also after end of cycling but at the same temperature and relative humidity used during cycling, for each test number in accordance with table XIII (see 4.7.6, 4.7.7, 4.7.8, 4.7.10, 4.7.16 and 4.7.17).

4.7.22.3 MTBF demonstration. From 4.7.22.1 data for all 24,000 cycles of operation, compute the MTBF assuming one cycle equals 1.0 hours.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Telephone systems and associated equipment. Telephone systems and associated equipment in the quantity specified (see 3.2.2) shall be packaged to protection level A or C, and packing level A, B or C as specified (see 6.2.1), and marked in accordance with MIL-E-17555.

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5.2 Sensitive electronic items.

5.2.1 Lead and terminal protection. Lead or terminal configurations shall be maintained as manufactured without causing electromagnetic pulses (EMP), electrostatic discharges (ESD) and electrical over-stress (EOS) of the item. Protection shall be by means of a carrier, container design or inserts of non-corrosive, electrostatic-free supporting materials. Materials used to maintain item position and lead or terminal configuration shall permit item removal and replacement without damage to the item.

5.2.1.1 Carrier. Carriers, when used for additional protection of miniature electronic items, shall be of such strength to prevent damaging resonances, shocks and electrostatic charges to the sensitive items. Anchoring or securing of the items, leads or terminals within the carrier by means of tape or adhesive shall be prohibited. The carrier shall maintain physical separation and manufactured configuration of the items, leads or terminals during packaging, handling, transportation, storage or stowage and for testing operations. The carrier shall permit safe and easy removal, inspection and item replacement and shall be designed without sharp edges to preclude subsequent damage to the item and packaging materials method.

5.2.2 Wraps and cushioning materials. Wraps and cushioning, when required for additional protection, shall be noncorrosive and in accordance with MIL-P-116, and shall not crumble, flake, powder or shed. Wraps or cushioning in direct contact with the electrostatic sensitive items shall be in accordance with the electrostatic protection requirements specified herein (see 5.2.3.1).

5.2.3 Unit protection. Unless otherwise specified (see 6.2.1), sensitive electronic items subject to degradation from electrostatic, electromagnetic forces, or both, shall be unit protected in accordance with methods IA or II of MIL-P-116 except as specified herein (see 5.2.3.1).

5.2.3.1 Electrostatic protection. Items adversely affected by electrostatic field forces shall be provided and initial wrap of material in accordance with MIL-B-81705, type II, or cushioned in material in accordance with PPP-C-1842, type III, style A or B, and unit packaged in heat-sealed bags in accordance with MIL-B-117, type I, class F, style 1. Alternatively, reclosable cushioned pouches in accordance with MIL-P-81997, type I or II may be used in lieu of the initial wrap or cushioning. Noncorrosive conductive materials shall be applied to all exposed leads and connector pins to maintain a common potential. This is to protect the items from electrostatic charges that may be encountered during handling.

5.2.4 Packaging materials. Packaging materials currently covered by title, scope or intended use under Government specifications, but modified as electrostatic-free materials, or newly developed electrostatic-free packaging materials not covered by a Government packaging material specification are encouraged for use. Use of such modified or newly developed electrostatic-free packaging materials will be permitted subject to the contracting officer's determination that the following applies: (a) The physical properties of such materials are equal to or better than similarly constructed materials covered under a required Government packaging material specification, (b) that such materials satisfy the electrostatic decay rate requirement of MIL-B-81705, and (c) that the proposed materials are in accordance with SE-003-AA-TRN-010 and DOD-HDBK-263

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requirements for protective material. The material manufacturer or contractor shall furnish to the Government inspector for review, documented proof of compliance with the requirements specified herein, certified by an acceptable testing laboratory. Upon submission for acceptance, copies of the contractor's document proof shall be forwarded to the contracting officer packaging activity. The decision of the contracting officer shall be final as to the acceptability or non-acceptability of the packaging material and the decision shall not be subject to review under the disputes clause of the contract. When such materials are acceptable, unit packaging shall be in accordance with the procedures for electrostatic protection.

5.2.5 Electromagnetic protection. Unless otherwise specified (see 6.2.1), items subject to damage by electromagnetic forces shall be unit packaged in heat-sealed, barrier bags in accordance with MIL-B-117, type I or II, class E, style 1 or type I, class F, style 1. When MIL-B-117, type I or II, class E, style 1 bags are selected and used, the barrier material shall also contain a laminate of aluminum foil and shall be in accordance with MIL-B-117.

5.2.6 Electromagnetic and electrostatic protection. When the items require both electromagnetic and electrostatic protection, unit packaging shall be as specified under electrostatic protection (see 5.2.5). In addition, guidance for identification of and control programs for electrical and electronic parts sensitive to electrostatic discharge shall be in accordance with DOD-STD-1686 and DOD-HDBK-263.

6. NOTES

6.1 Intended use. The sound-powered telephone system is intended for use as a shipboard interior voice communications system. When power is lost this system is expected to perform, so simplicity and ruggedness is required.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Quantities and types of each system component (see 1.1 and 3.2.2).
(c) If first article is required (see 3.1).
(d) Quantities of H-203/U headset and handset holder if other than specified (see 3.6.1.3).
(e) Whether terminal stations require front crank or side crank (see 3.6.1.6.9).
(f) Whether the equipment shall be energized and operated at rated signal voltage and load current, as specified (see 3.9).
(g) Whether a R/M prediction plan is required (see 3.10).
(h) Preservation, packaging, packing and marking if other than specified (see 5.1, 5.2.3 and 5.2.5).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with

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the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.10	Report, failure summary and analysis	DI-R-7041	
	Nonstandard part approval requests/proposed additions to an approved program parts selection list	DI-E-7028	
	Reliability prediction report	DI-R-7082	
	Report, maintainability prediction	UDI-R-23567	
3.11	Manual, technical, preliminary	DI-M-2043	MIL-M-15071, type II
	Manual, technical, standard	DI-M-2044	
	Reproducible, camera ready copy	UDI-A-23020	
4.1.1	Inspection system program plan	DI-R-4803	
4.4.1	Procedures, test	UDI-T-23732	
4.4.1.1	Notification of tests	DI-T-23731	
4.4.2	Report, first article test	UDI-T-23790	
4.6.1	Report material inspection system	UDI-T-23741	

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such

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products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

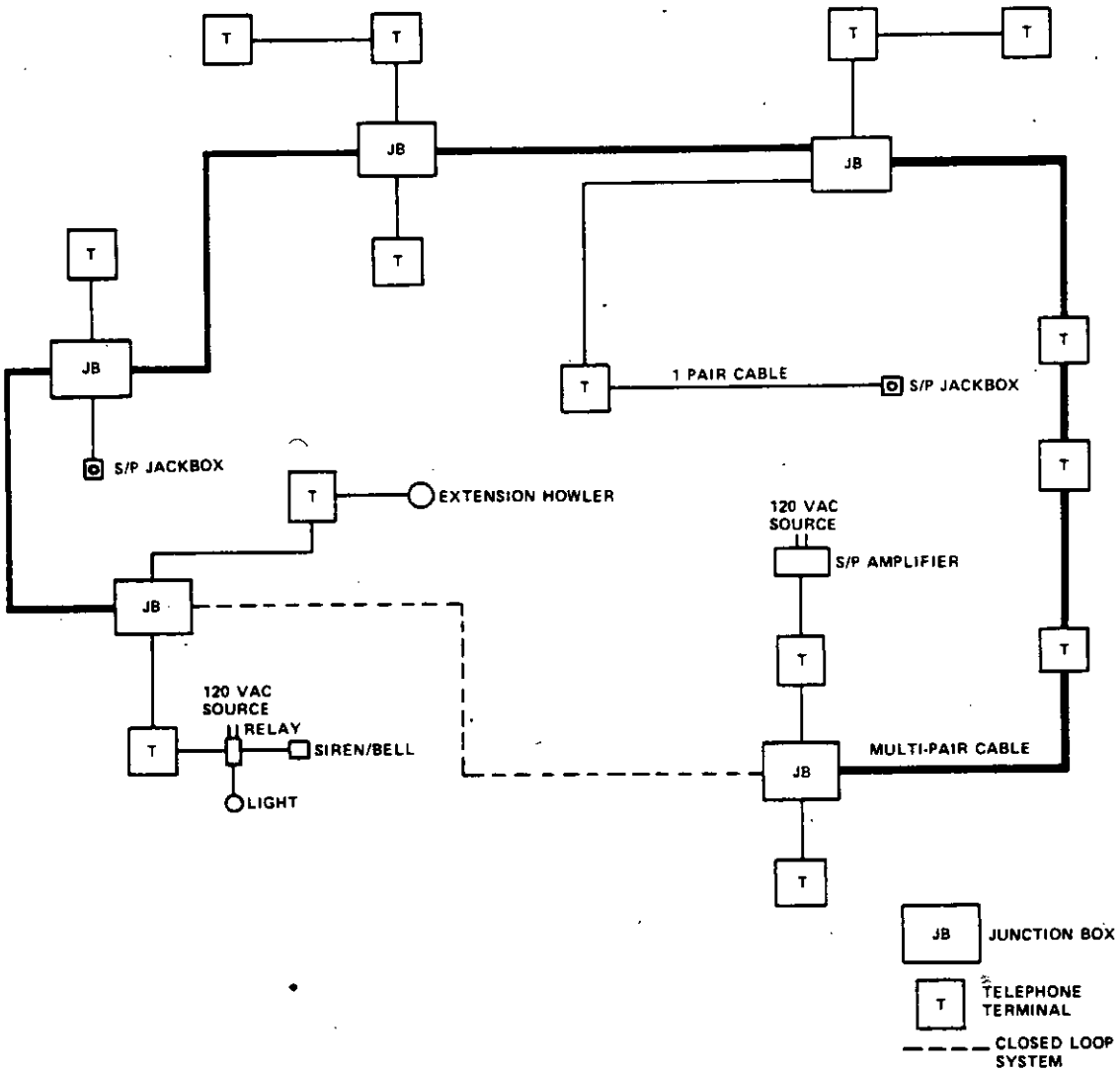
6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.5.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

Preparing activity:
Navy - SH
(Project 5965-N180)

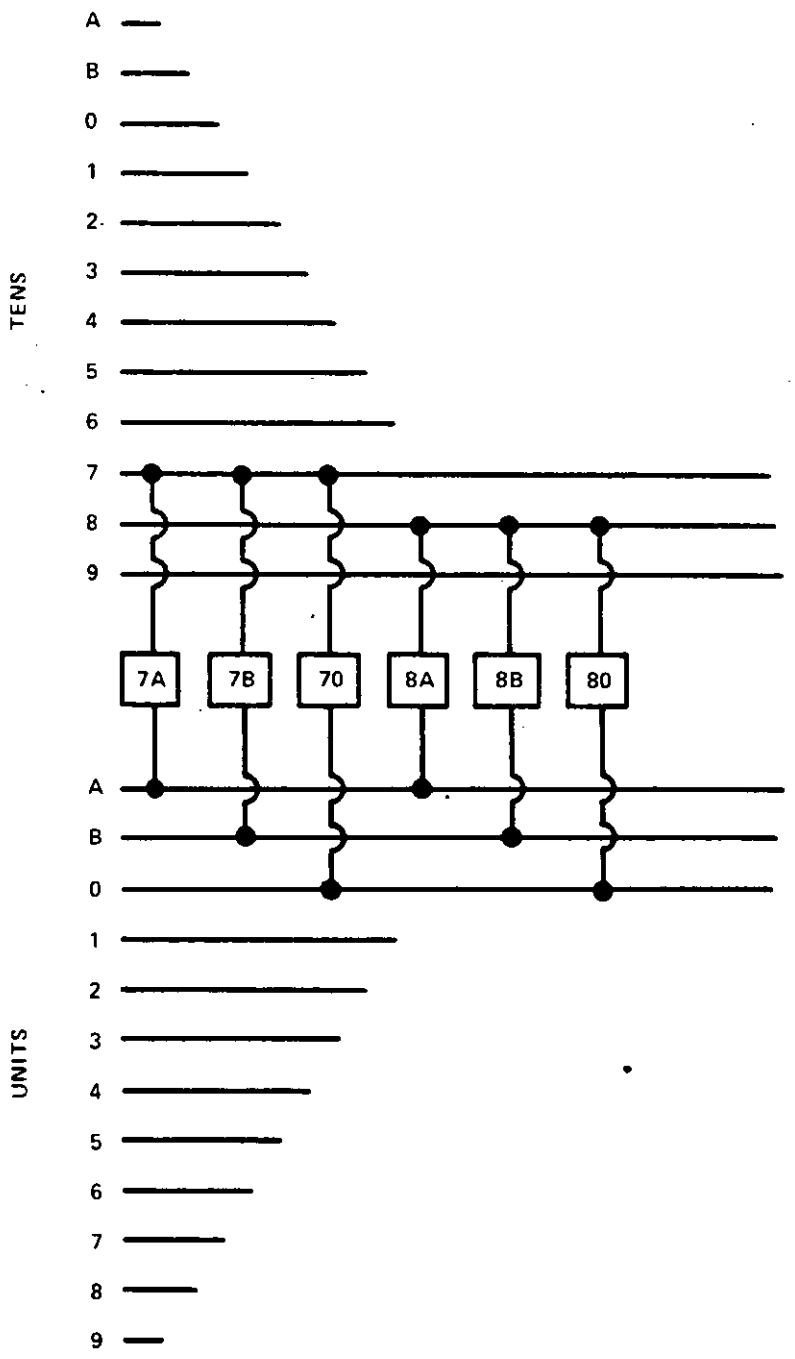
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SH 12526

FIGURE 1. Simplified system configurations of AN/WTC-2(V).

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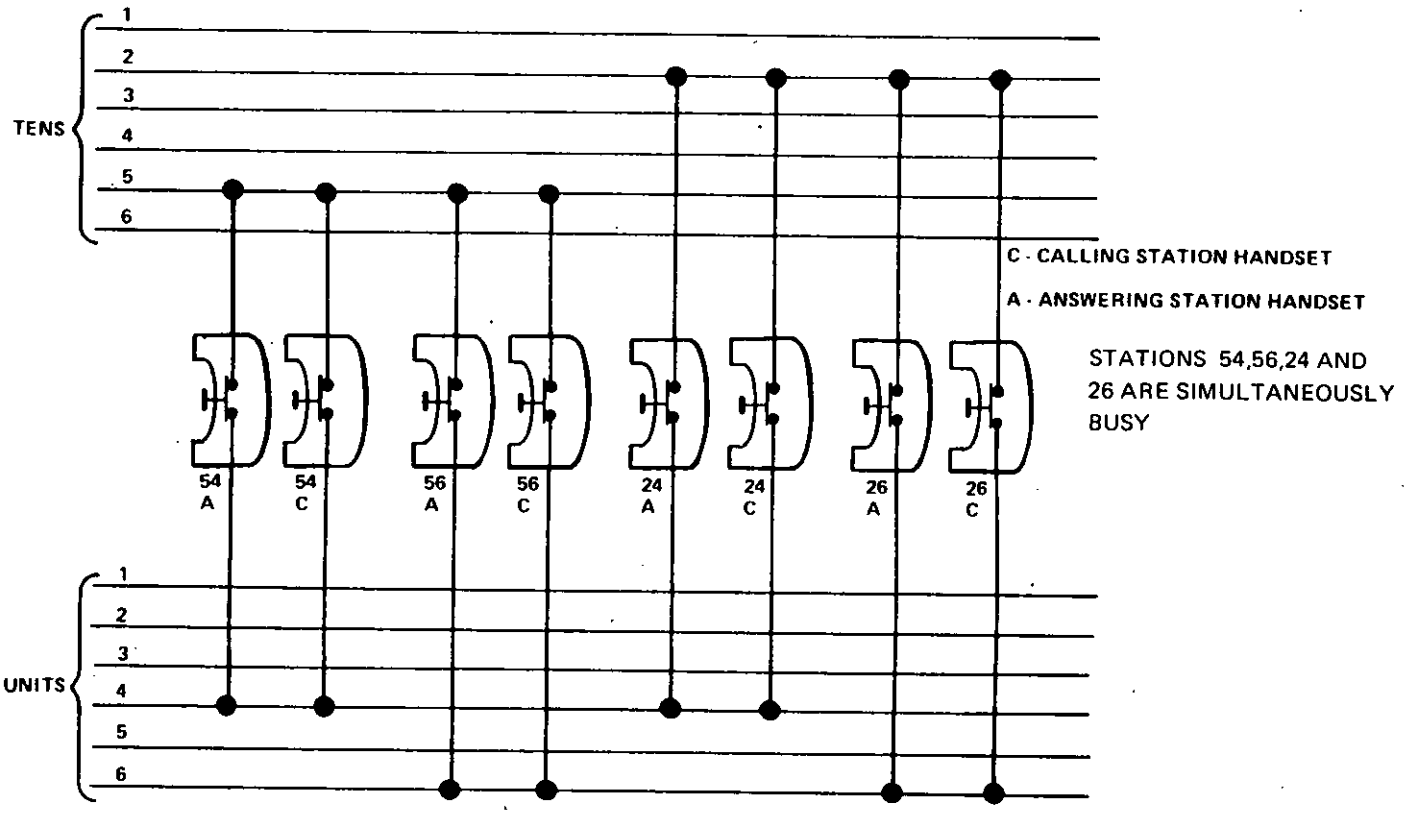


SH 12527

When station 80 is called, the ring and talk voltages are impressed on "8" wire in the tens group, and the "0" wire in the units group. Identically numbered wires in the tens and units group constitute a twisted pair in the system interconnecting cable (see table II).

FIGURE 2. Line sharing illustration.

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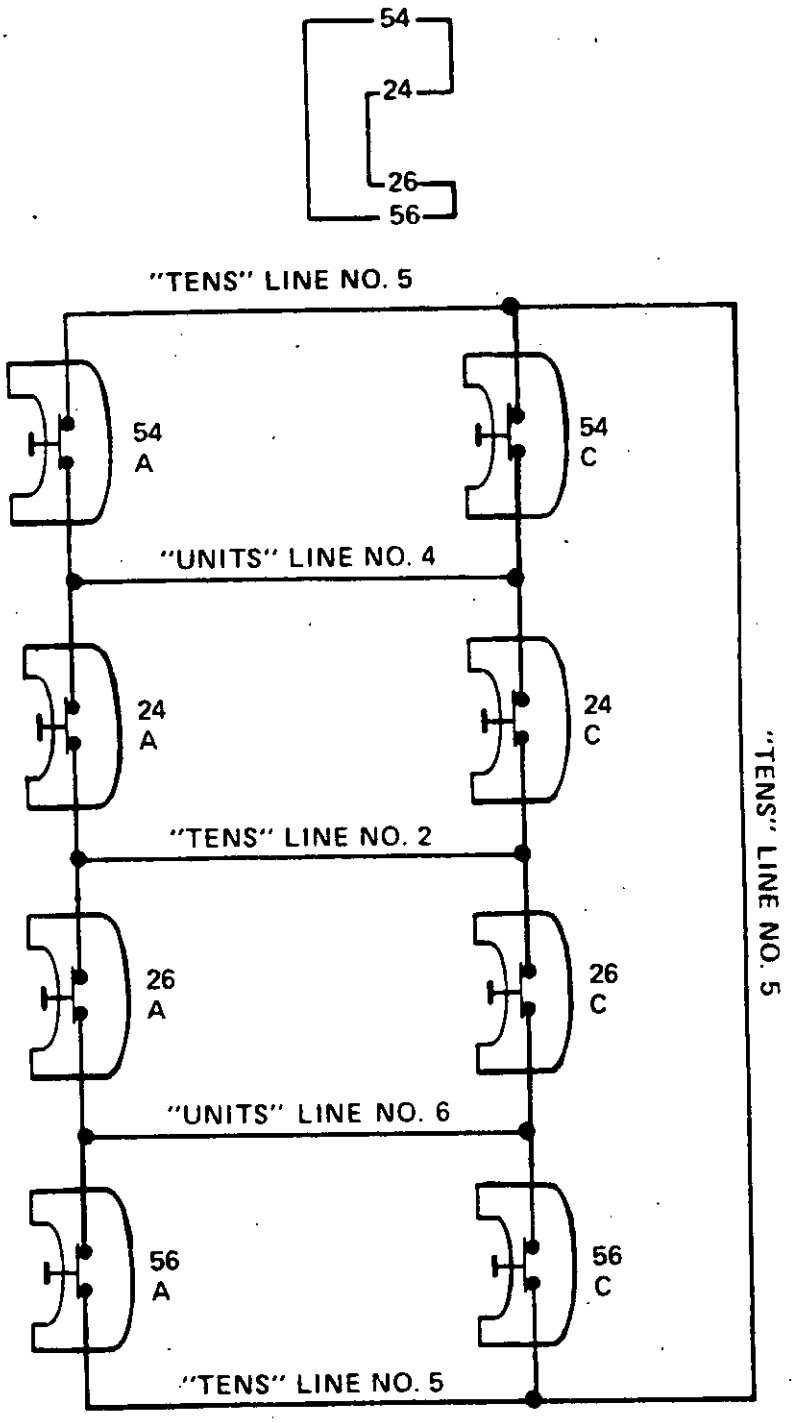


SH 12528

Looping is an overload condition resulting from certain simultaneous station-to-station calls being made in the system; it is the undesirable and unintentional formation of a complex series-parallel circuit among stations that are simultaneously busy. Occurrence of looping is minimized by use of the number assignment tables.

FIGURE 3. Illustration of looping.

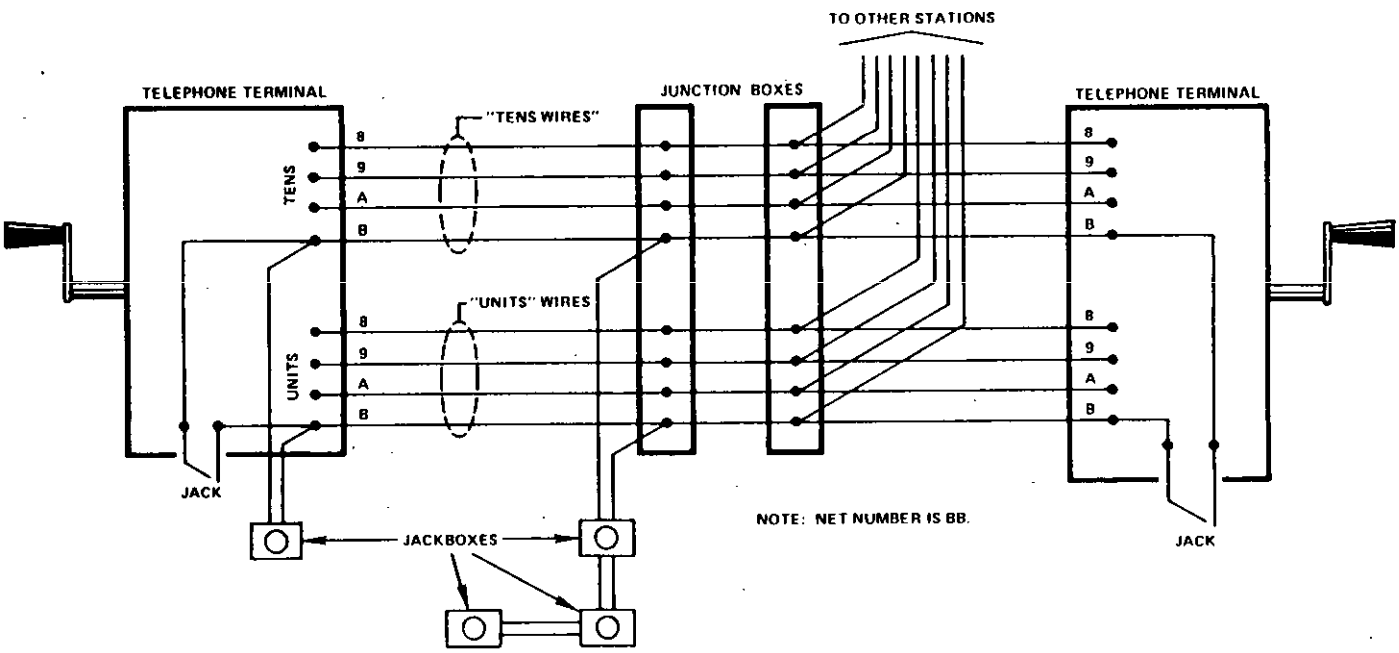
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SH 12529

FIGURE 4. Equivalent looping circuit (equivalent to figure 10).

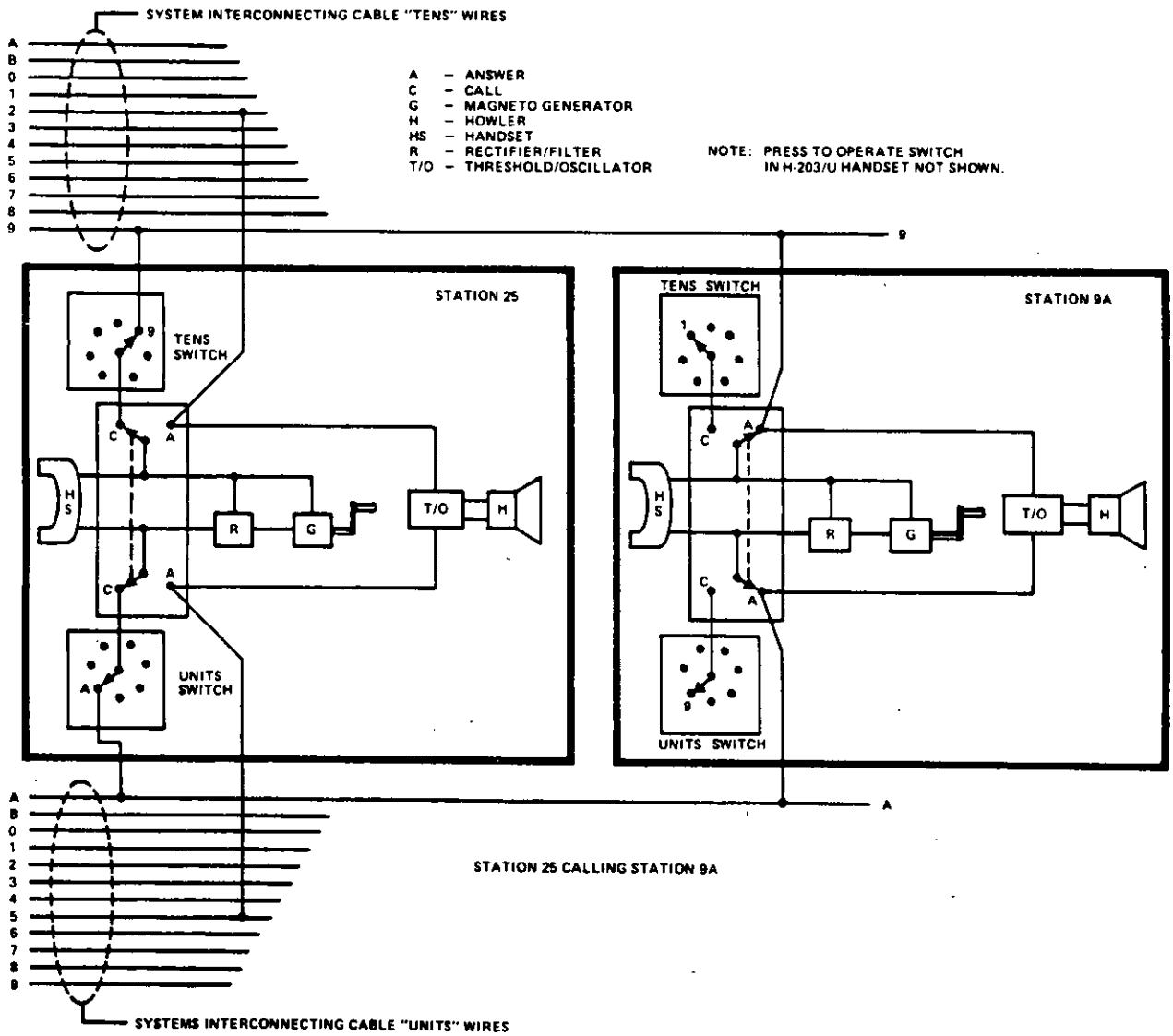
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SH 12530

FIGURE 5. Example integrated station-to-station and net communication system (simplified diagram).

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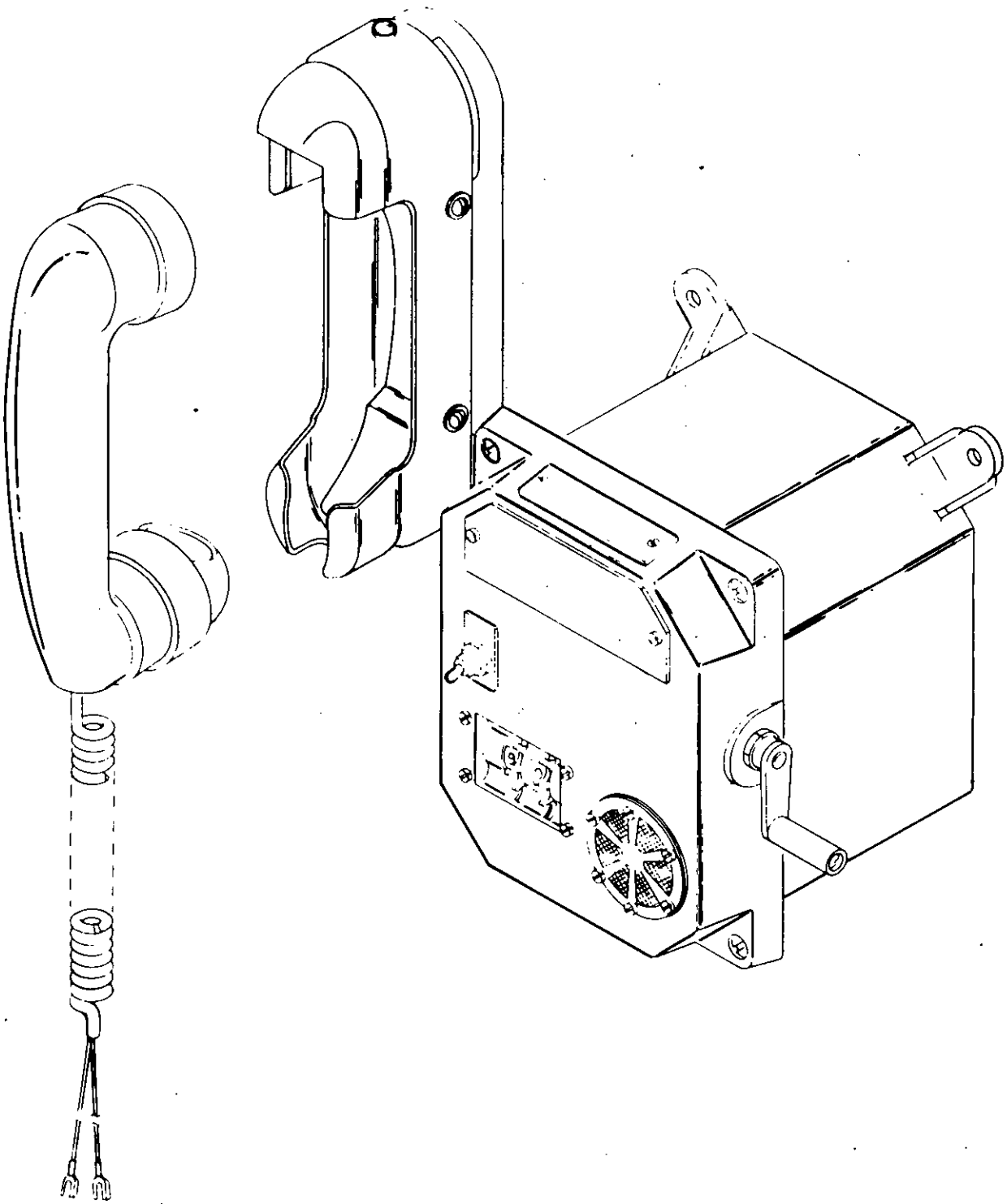


SH 12531

Example electrical interconnections for making a station-to-station call, showing station 25 calling station 9A. A ring signal from station 25 is received at station 9A over tens wire 9 and units wire A, activating the howler at station 9A. Communications then takes place over the same wires using the handsets. Note that station 25 is in the CALL mode, and station 9A is in the ANSWER mode.

FIGURE 6. Example electrical interconnections for making a station-to-station call.

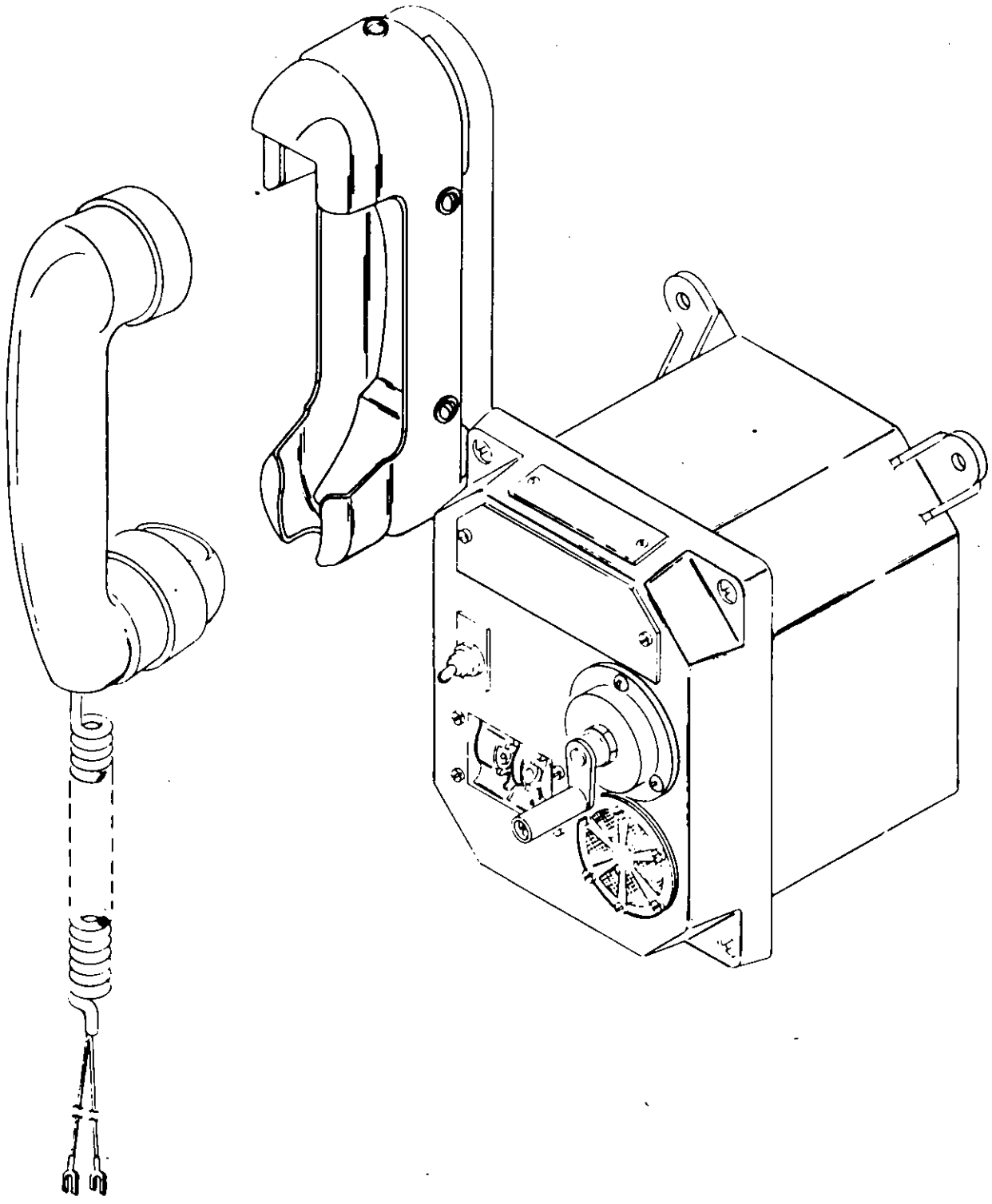
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SH 12532

FIGURE 7. Side crank, sound powered telephone set TA-974/WTC-2(V).

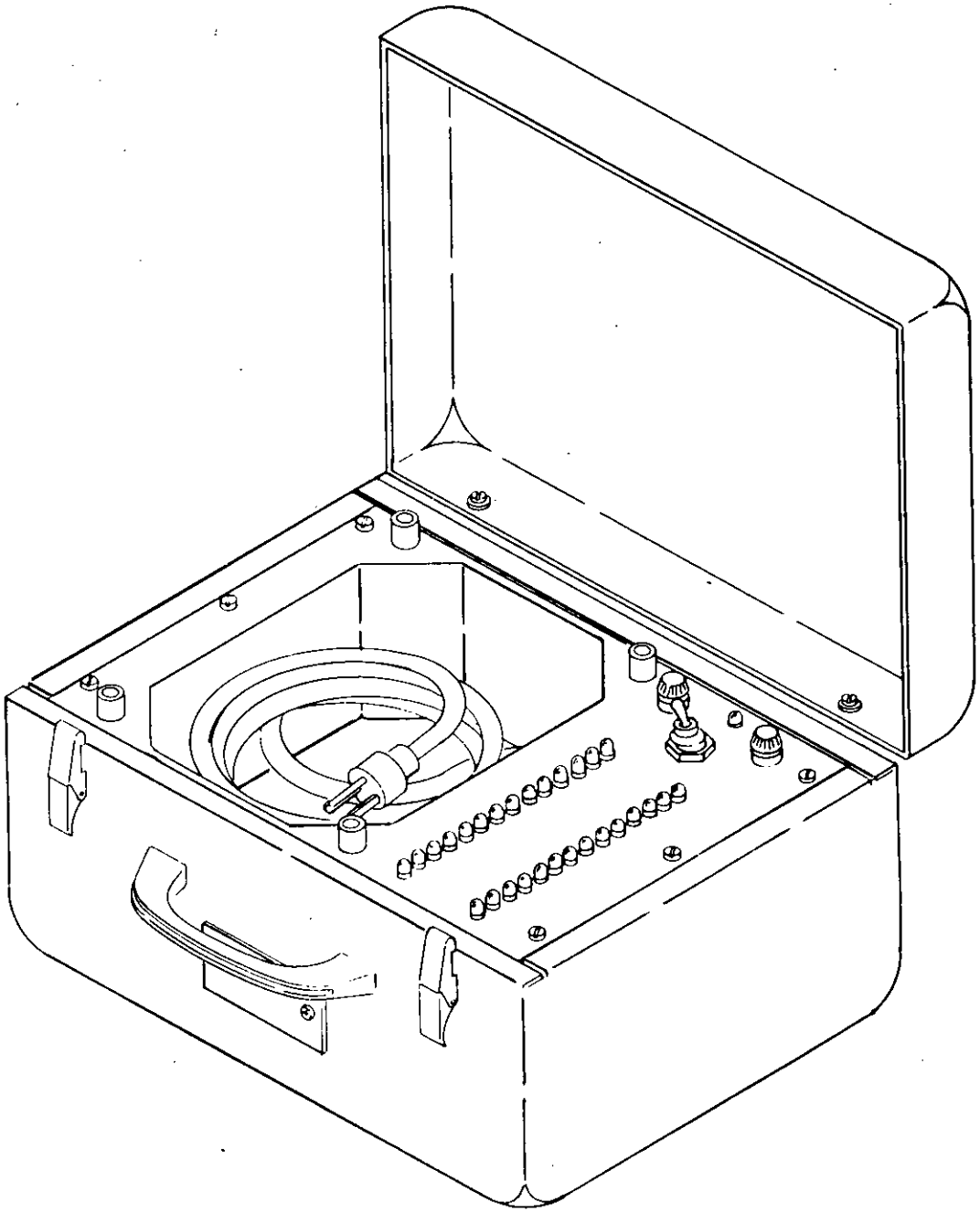
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SH 12533

FIGURE 8. Front crank sound powered telephone set TA-975/WTC-2(V).

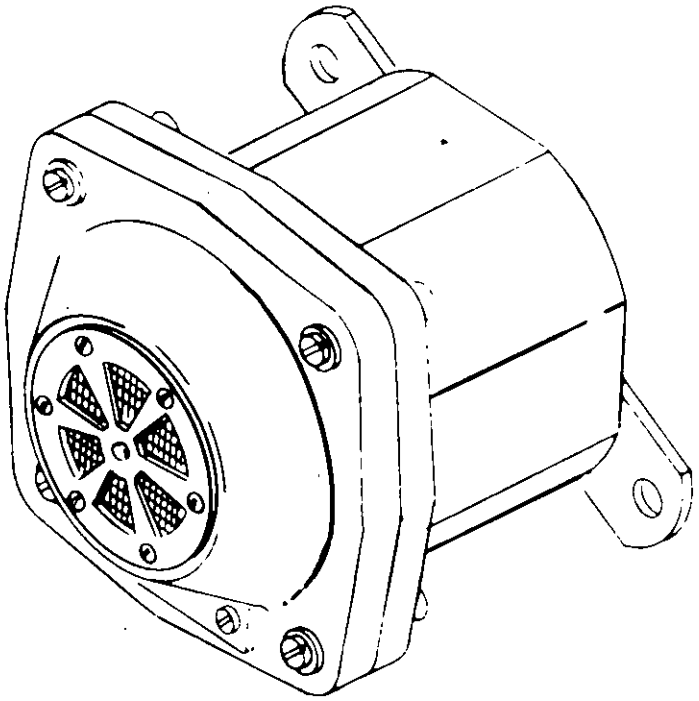
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SH 12534

FIGURE 9. Telephone test set TS-3687/WTC-2(V).

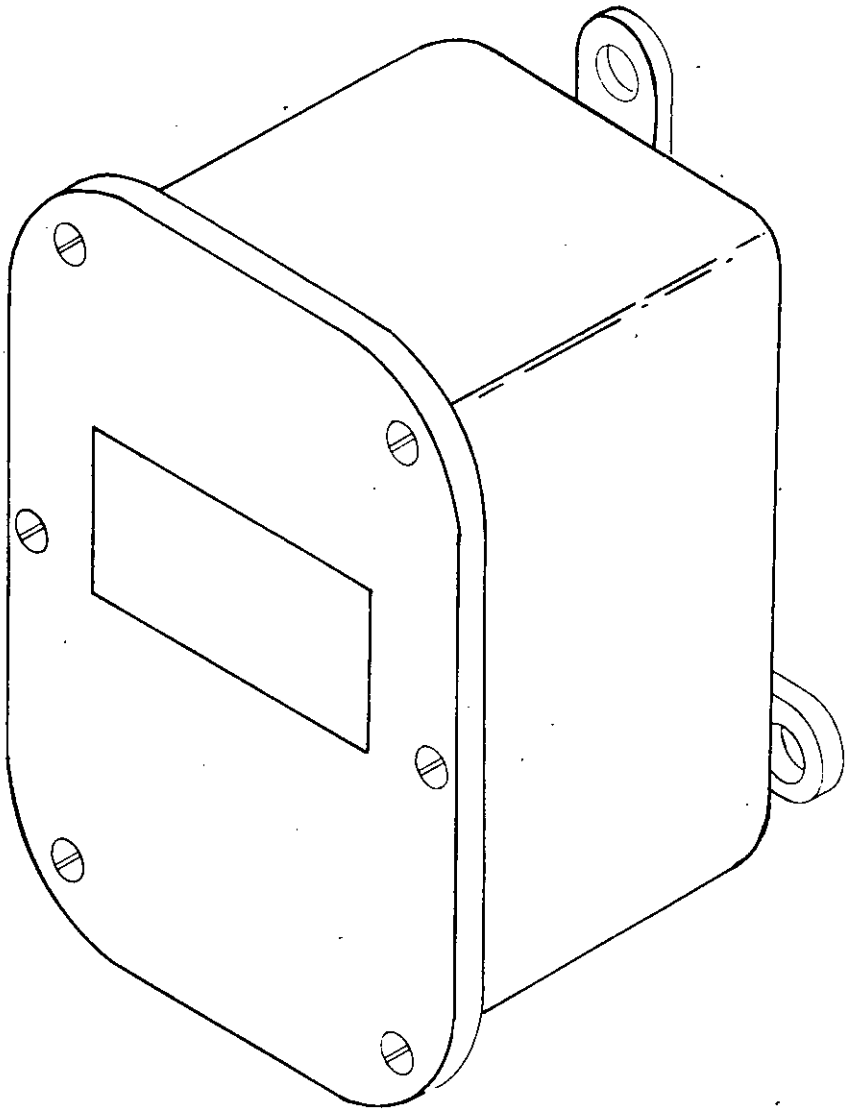
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SH 12535

FIGURE 10. Audible alarm BZ-240/WTC-2(V).

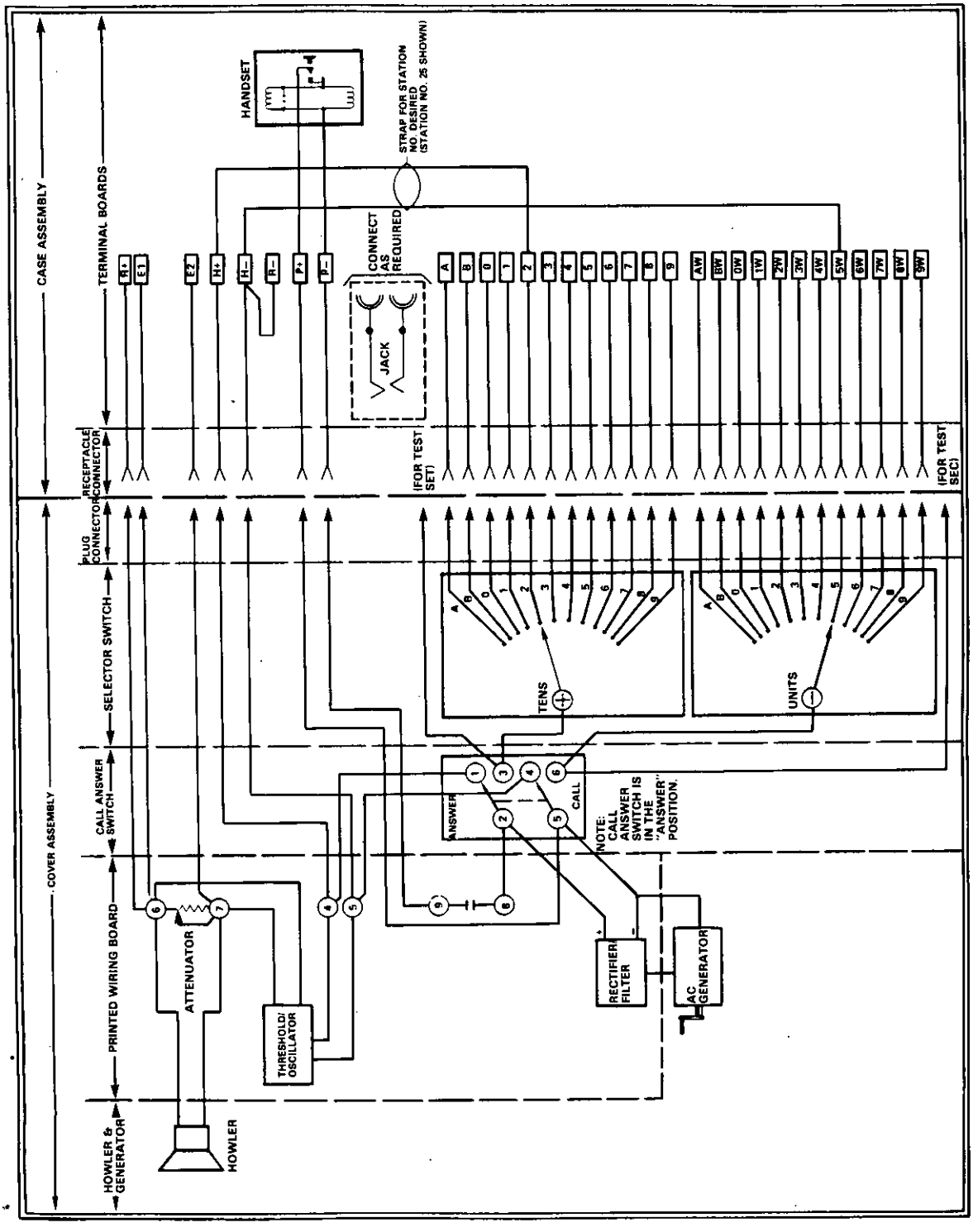
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SH 12536

FIGURE 11. Distribution box J-3523/WTC-2(V).

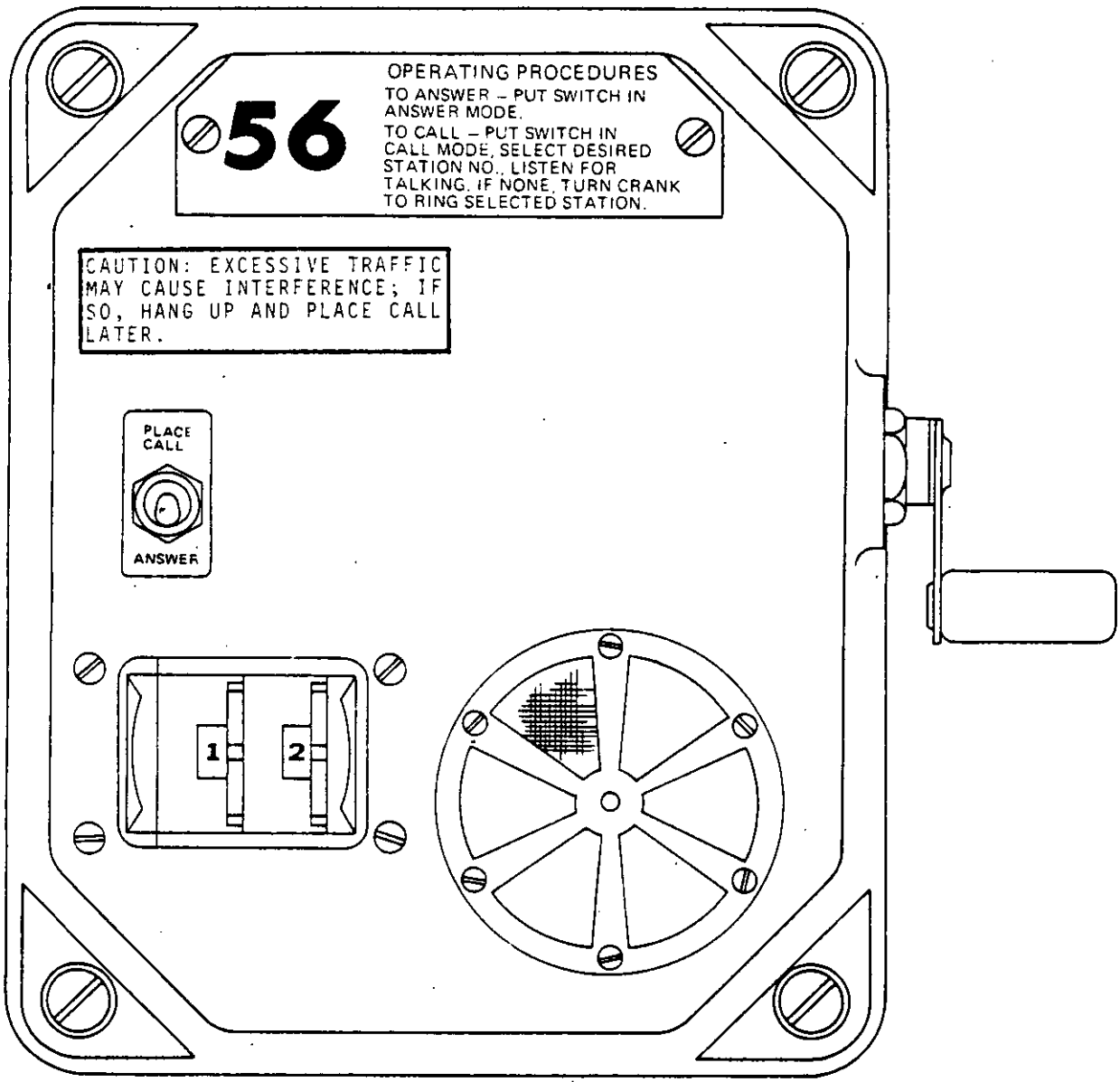
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SH 12537

FIGURE 12. Telephone terminal - basic schematic diagram.

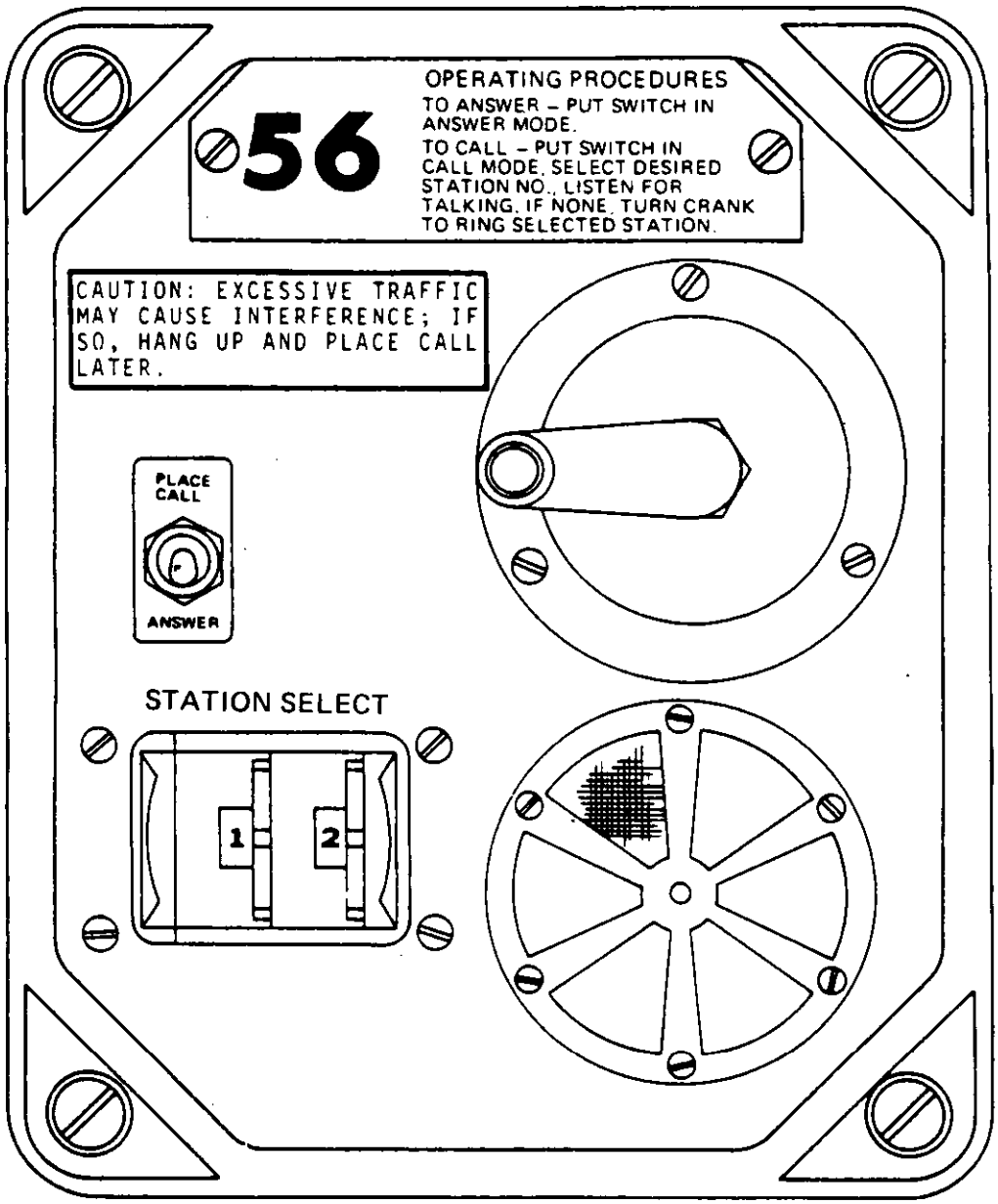
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SH 12538

FIGURE 13. Call signal station cover - side-crank model.

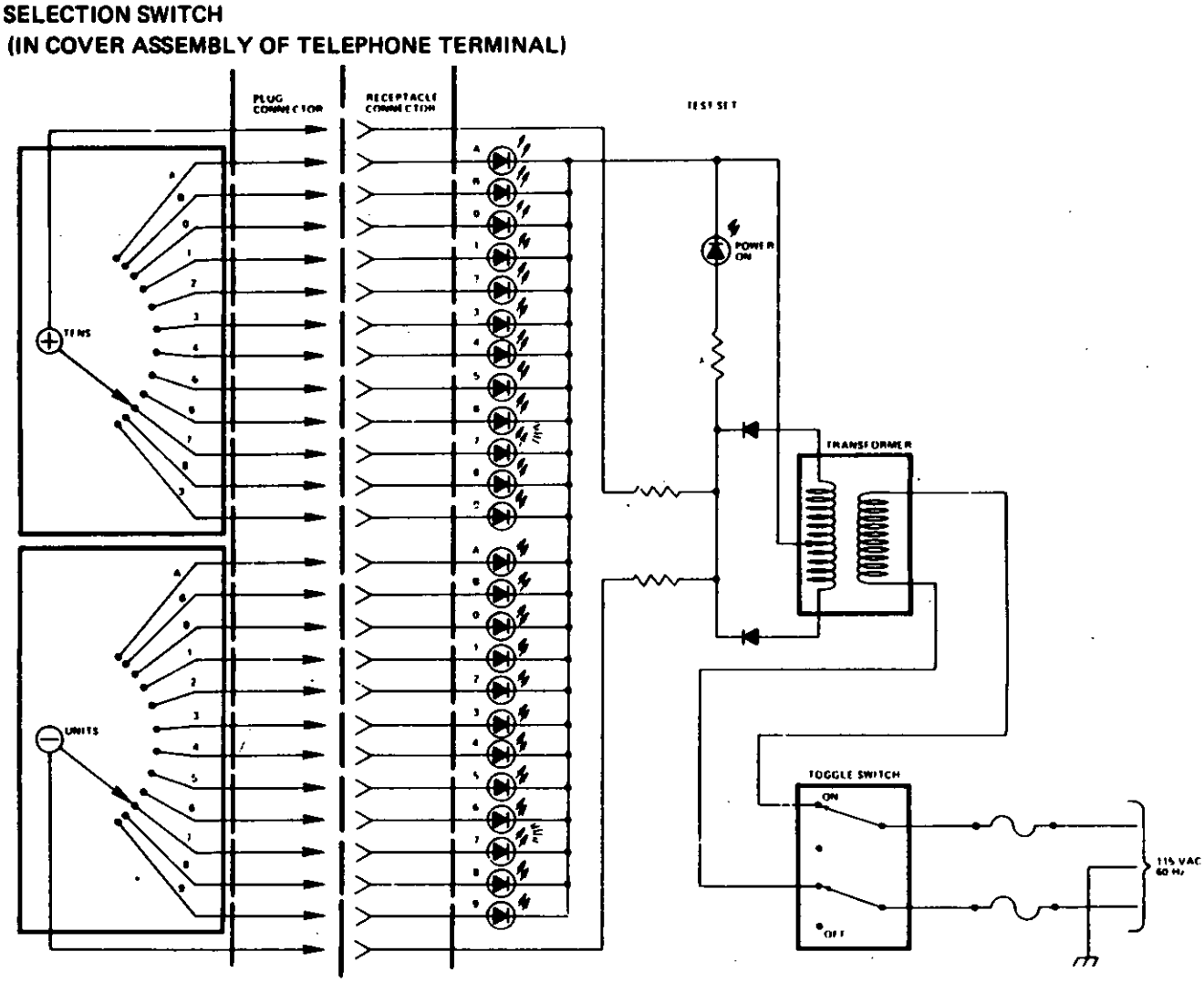
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SH 12539

FIGURE 14. Call signal station cover - front-crank model.

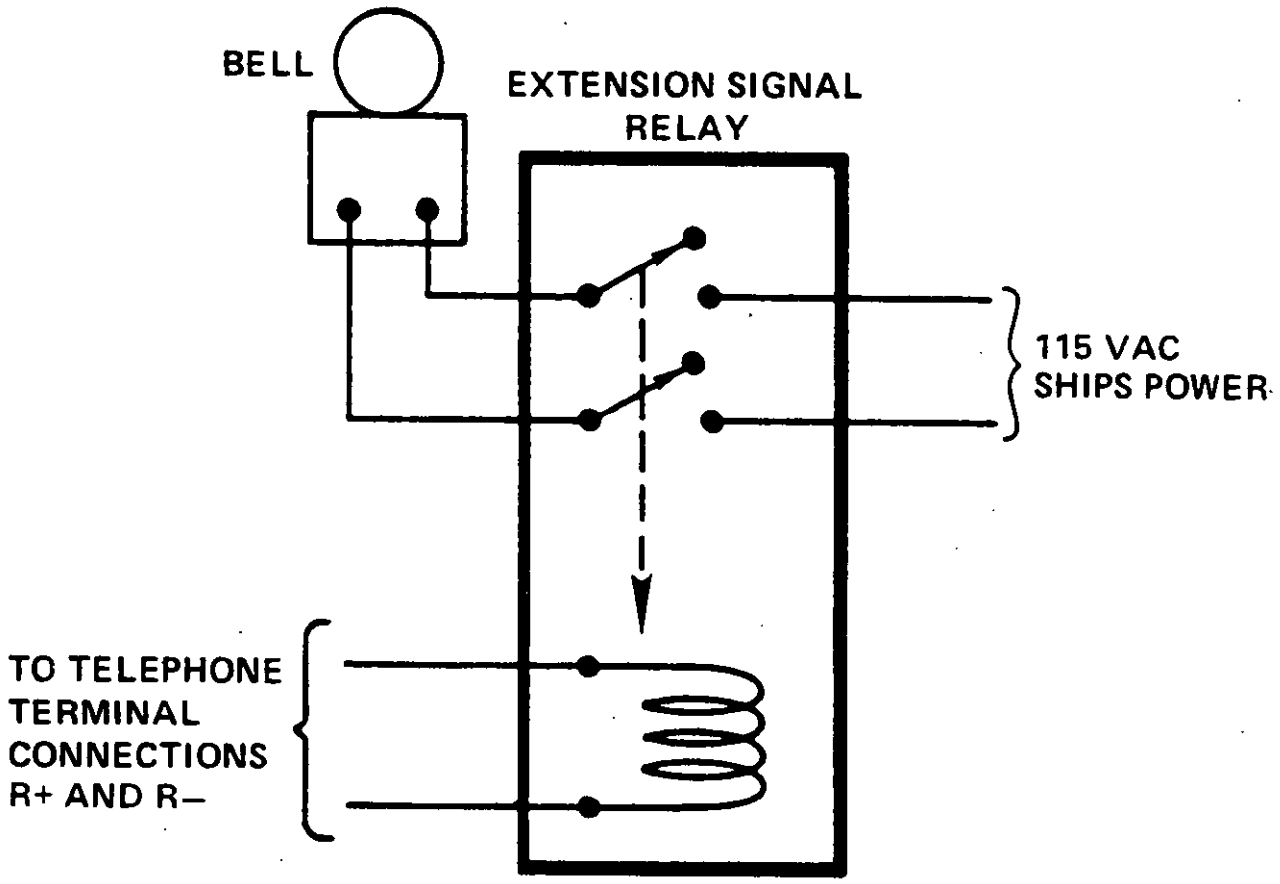
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SH 12540

FIGURE 15. Test set-telephone terminal cover assembly interconnection diagram.

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SH 12541

FIGURE 16. Distribution box (extension signal relay) schematic diagram.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-24632(SH)	TELEPHONE SYSTEM, SOUND-POWERED, SHIPS SERVICE AN/WTC-2()(V)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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